

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df(data):
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    # Removing of the columns where there is more than 15% of nan
    #df_features = pd.DataFrame(features_all)
    cols_to_check = target_col + features_X
    nan_ratio = df[cols_to_check].isna().mean()
    valid_features = nan_ratio[nan_ratio <= 0.15].index
    valid_features_list = list(valid_features)
```

```

#df = df[valid_features]
# Removing of last np.nan
mask = np.isfinite(df[valid_features]).all(axis=1)
df = df[mask]

cols = list(dict.fromkeys(valid_features_list + #target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

new_feats_X = [x for x in features_X if x in valid_features_list]
new_feats_add = [x for x in feats_eng if x in valid_features_list]

data["df"]=df
data["features_X"] = new_feats_X
data["feats_added"] = new_feats_add

column_removed = [x for x in cols_to_check if x not in valid_features_list]
print(f'Column removed because np.isnan > 15%: {column_removed}')

return data

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()

```

```

        with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r"):
        #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

df["hot"] = df.apply(assign_hot_season,axis=1)
df["umidity_range"] = df.apply(assign_humidity, axis = 1)
#Removing of rabbit 4
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}

```

```

elif camp == "2026_Gen":
    #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
    #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
    mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
else:
    mapping = {}

return mapping.get(rid, "no")

```

Column removed because np.isnan > 15%: []

1.2 Metadata

Nombre de lignes df	8168
Nombre de lignes df clean	8168
Nombre de features	5
% de NaN	0.0
Taille dataset entraînement	5717
Taille dataset test	2450
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Model(s) analysis

2.1 Explanation legend tables

Columns

- rank: ranking regarding cross validation test

- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_test < 0.95$ (under performance of the test)

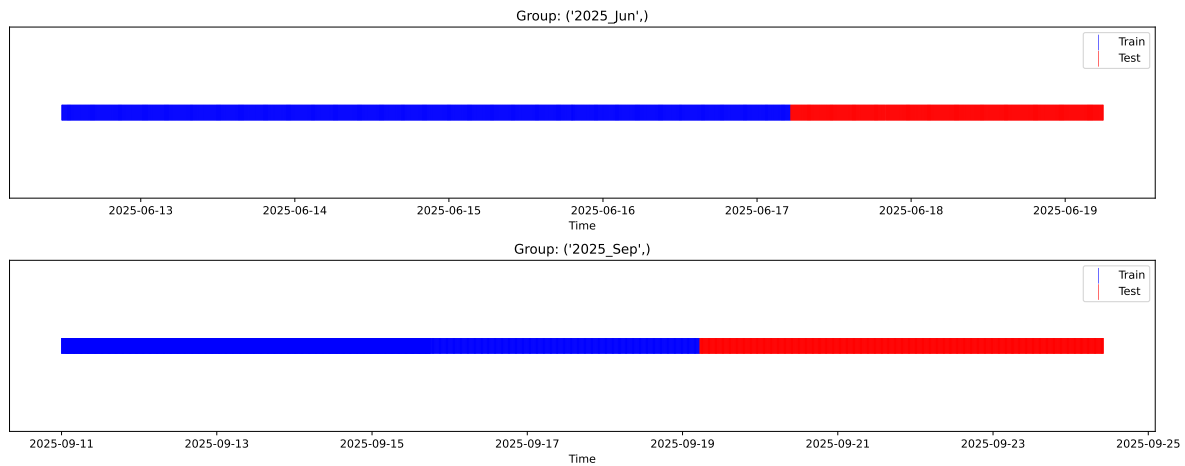
[Green] $r_test > 1.05$ (over performance of the test)

[Red] r_train not in $[0.95, 1.05]$ (gap between train/test)

2.2 Repartition of train and test

2.2.1 External Split: X_train vs X_test

External Split: Train: 5719 samples Test: 2449 samples



2.2.2 Internal CV Folds

Fold 1/5

Train: 4003 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



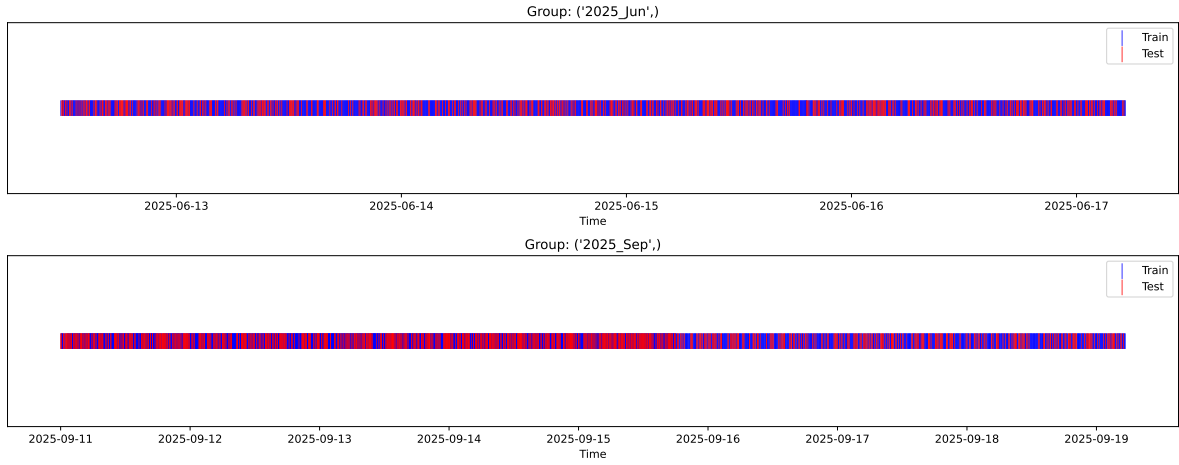
Fold 2/5

Train: 4003 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



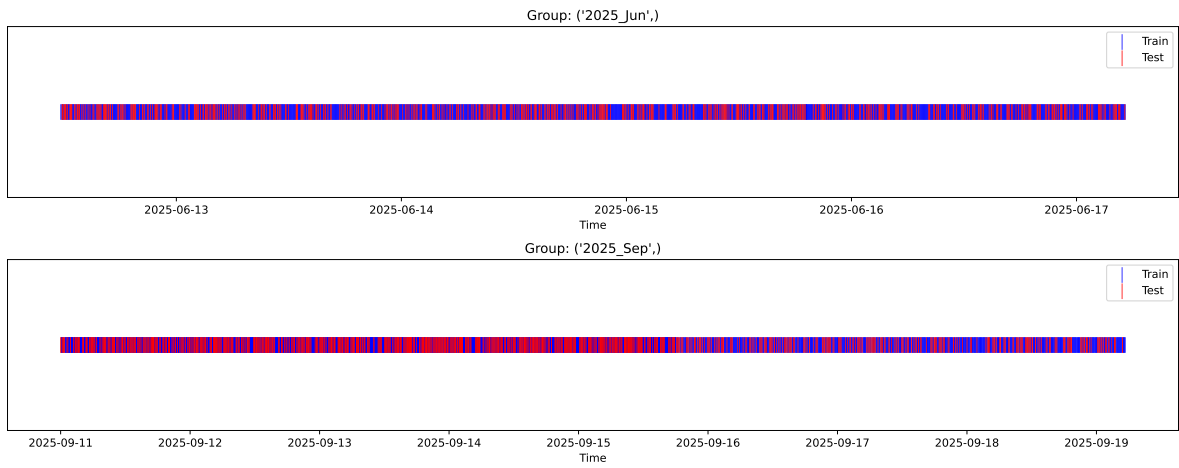
Fold 3/5

Train: 4003 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



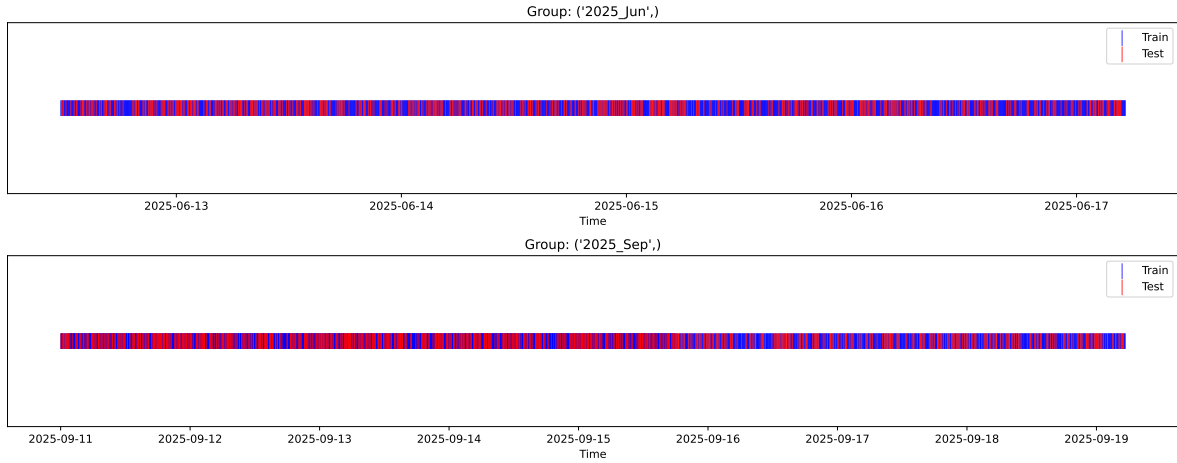
Fold 4/5

Train: 4003 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 4003 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



2.3 Model : Linear

2.3.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.354, std=0.804, min=0.959, max=8.682
 y_{test} : mean=2.340, std=0.772, min=1.071, max=13.845

Fold 0 — y_{test} : mean=2.35, std=0.79, y_{train} : mean=2.35, std=0.81

Fold 1 — y_{test} : mean=2.37, std=0.79, y_{train} : mean=2.35, std=0.81

Fold 2 — y_{test} : mean=2.36, std=0.80, y_{train} : mean=2.35, std=0.81

Fold 3 — y_{test} : mean=2.33, std=0.82, y_{train} : mean=2.36, std=0.80

Fold 4 — y_{test} : mean=2.33, std=0.77, y_{train} : mean=2.36, std=0.82

train scores CV: [0.40125516 0.40930372 0.40130967 0.40621486 0.39718676]

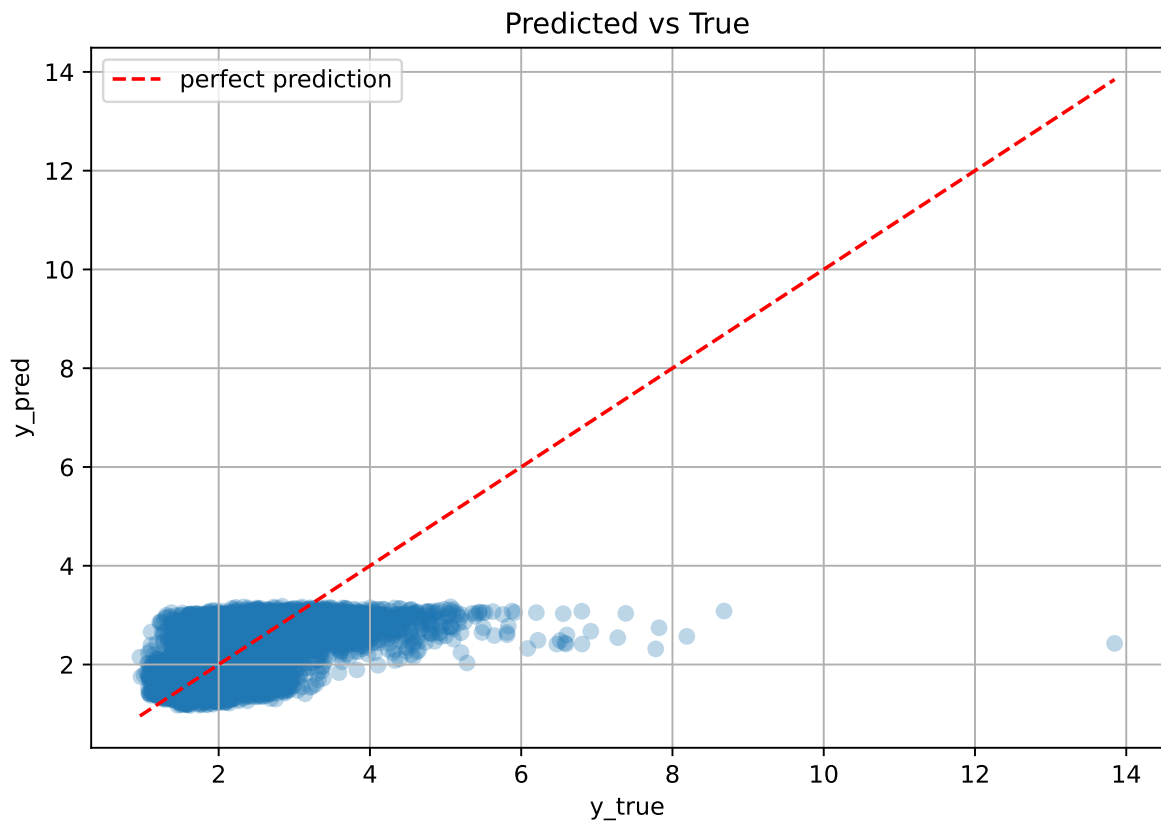
test scores CV: [0.41310399 0.39293914 0.41249368 0.39999369 0.42378683]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.4031	0.4085	0.0108	0.1872	0.1866	-0.2219	0.4568	0.9868

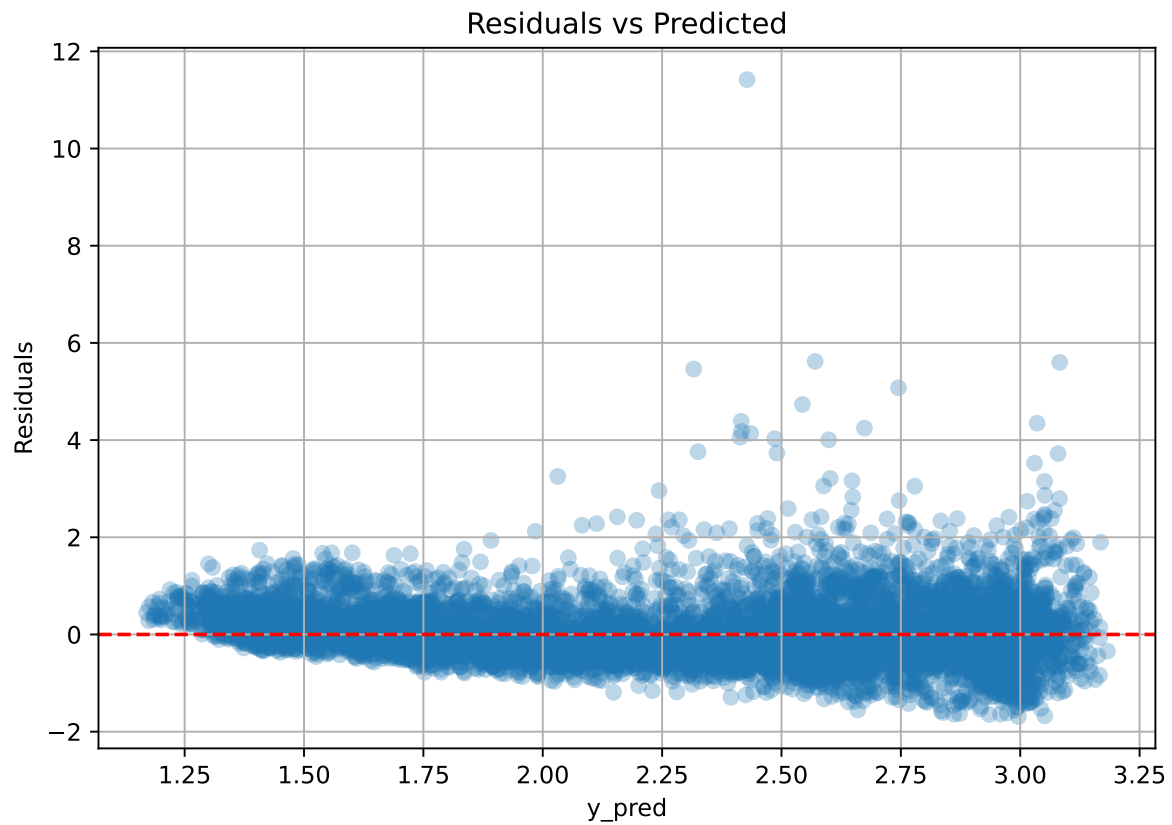
Coefficients:

coef_r_CO2 = -0.0195, coef_r_NH3 = -0.0421, coef_r_THI = 0.8947, coef_r_temperature = -1.1610, coef_r_umidity = -0.8420, intercept = 2.3535,

2.3.2 Scatter



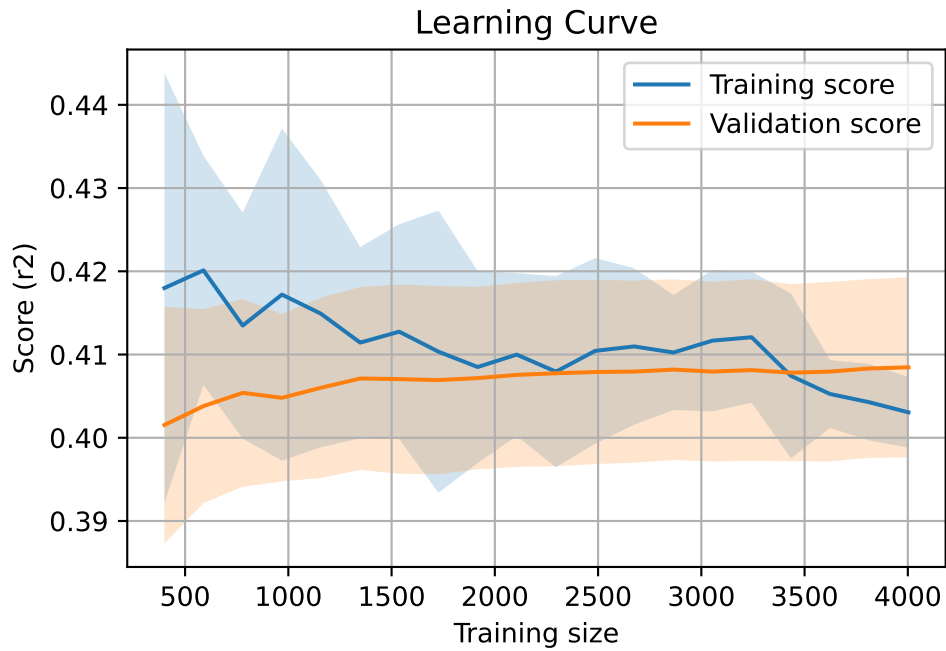
2.3.3 Scatter residuals



2.3.4 Learning curve

Len(group) : 8168, **Len(training)** : 5718,

Len(training_real) : 4003, **Len(test_of_training)** : 1715, **Len(test)** : 2450

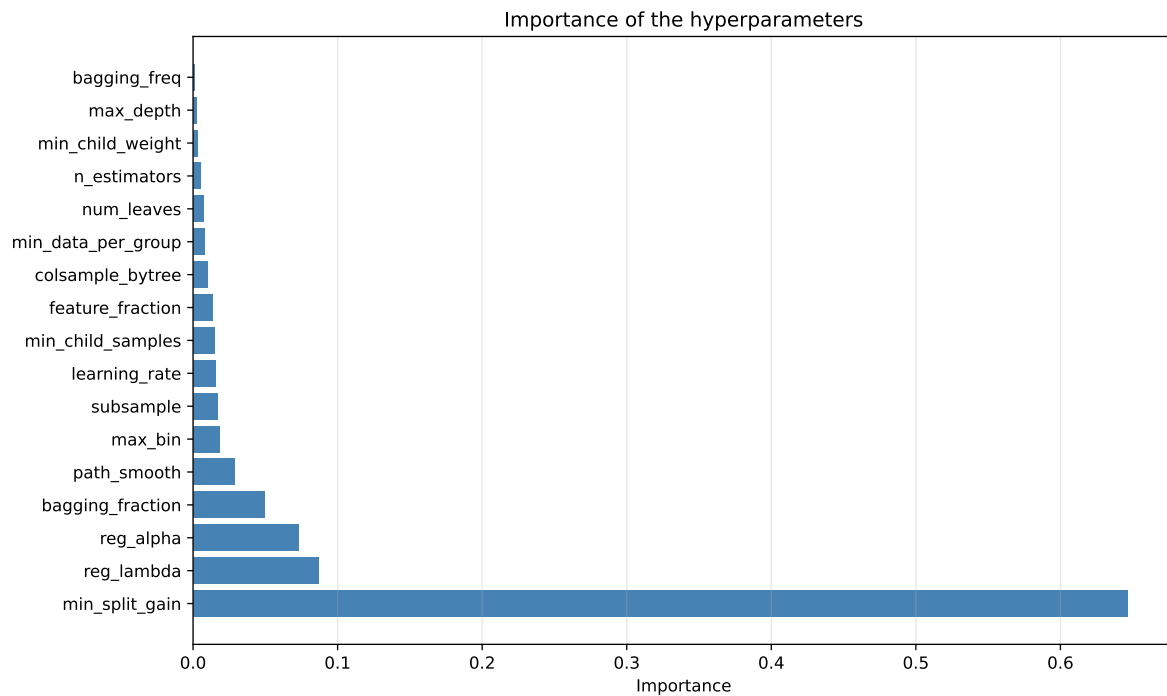
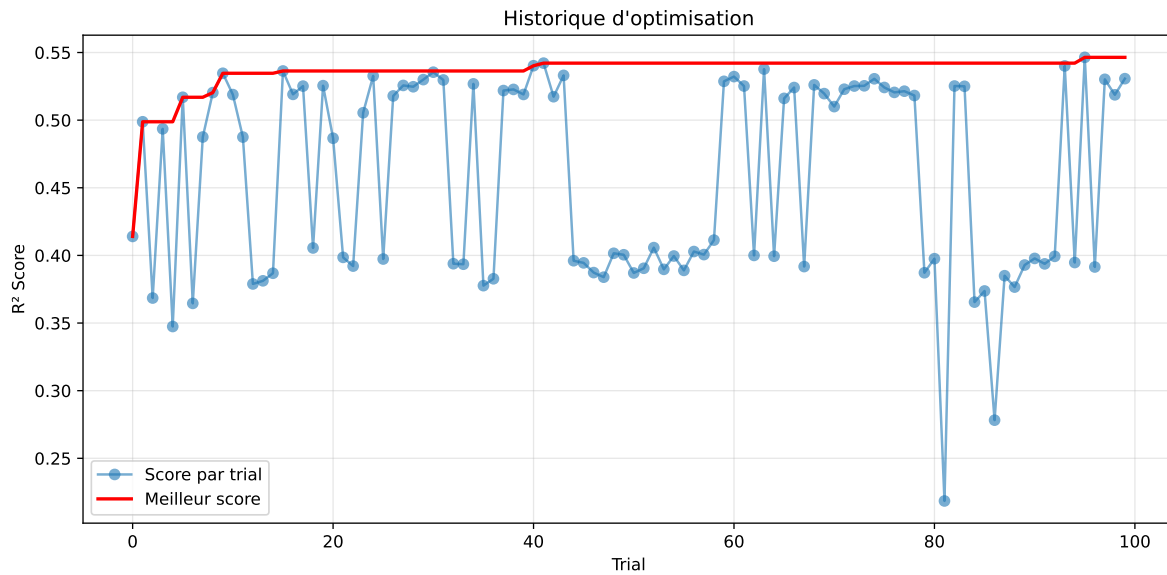


2.4 Model : LGBM

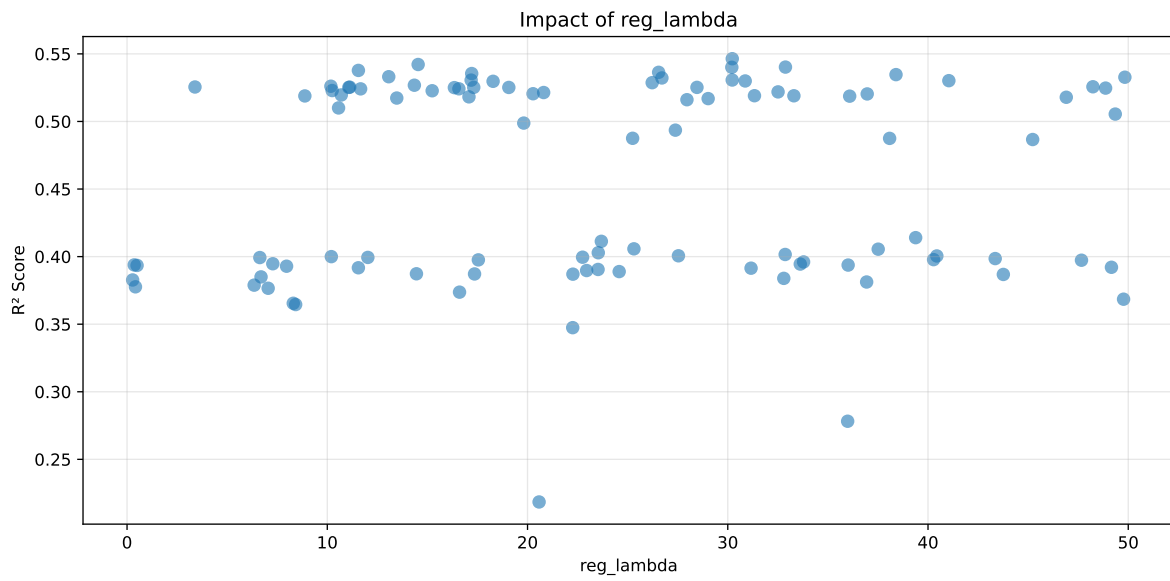
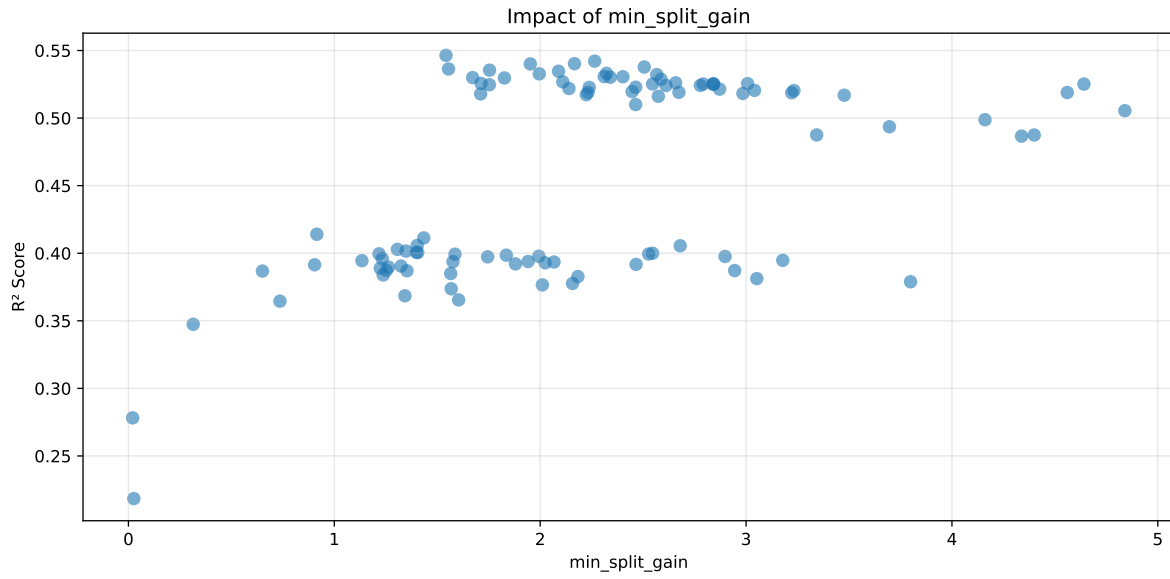
2.4.1 Gridsearch

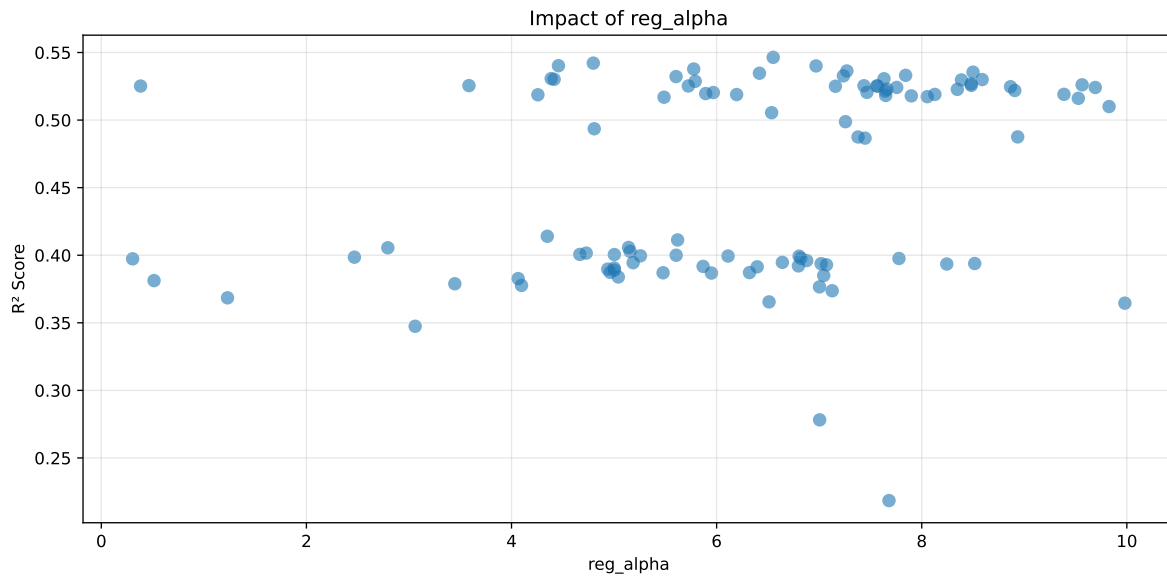
Distribution y_train vs y_test: y_train: mean=2.354, std=0.804, min=0.959, max=8.682
y_test: mean=2.340, std=0.772, min=1.071, max=13.845

===== TOP Importance FEATURES =====
feature importance r_CO2 550 r_umidity 385 r_temperature 316 r_NH3 303 r_THI 255



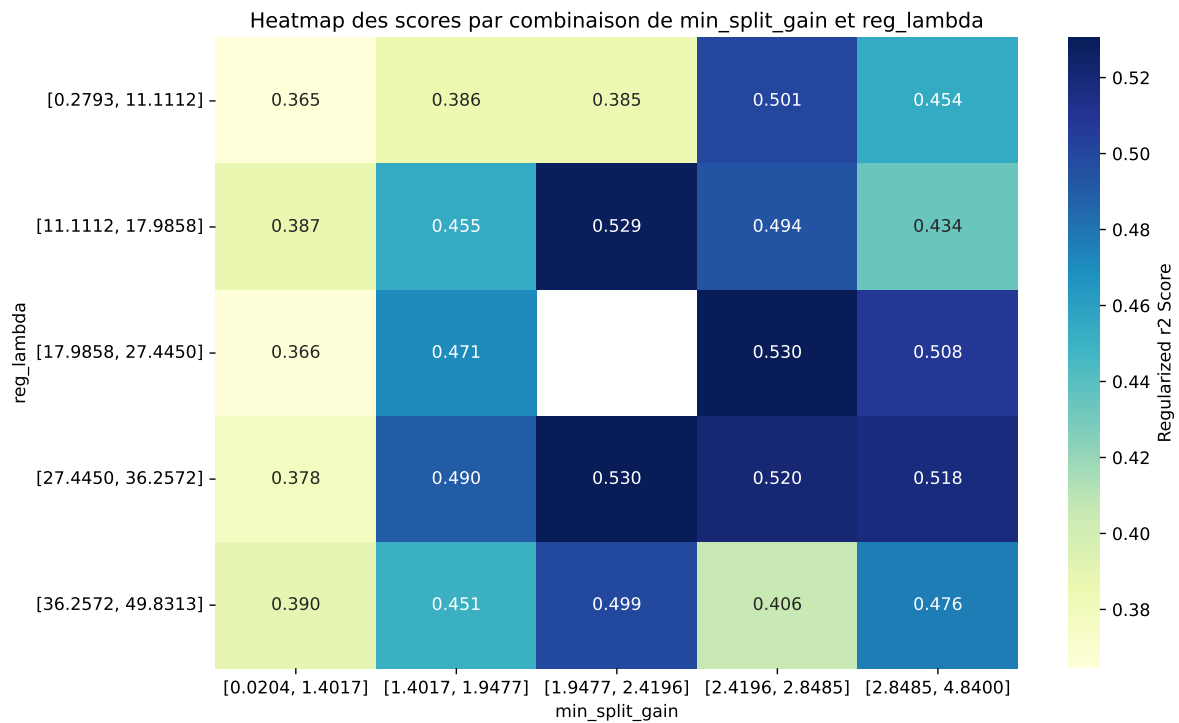
Paramètres avec >5% d'importance : ['min_split_gain', 'reg_lambda', 'reg_alpha']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5732	0.5464	0.2792	0.0182	0.5464	0.2731	-0.2733	0.4999	1.049
2	0.5689	0.5421	0.2834	0.0158	0.5421	0.2724	-0.2697	0.5025	1.0493
3	0.5656	0.5402	0.2788	0.0171	0.5402	0.2694	-0.2708	0.4987	1.047
4	0.5661	0.5401	0.2786	0.0208	0.5401	0.2746	-0.2655	0.5084	1.0481
5	0.5646	0.5378	0.2843	0.0184	0.5378	0.2809	-0.2569	0.5224	1.0499
6	0.563	0.5364	0.2785	0.0225	0.5364	0.274	-0.2624	0.5108	1.0498
7	0.5609	0.5355	0.2798	0.0175	0.5355	0.2768	-0.2586	0.517	1.0475
8	0.5595	0.5347	0.2771	0.0175	0.5347	0.2725	-0.2621	0.5097	1.0465
9	0.5584	0.5331	0.284	0.0168	0.5331	0.2773	-0.2558	0.5201	1.0475
10	0.5582	0.5327	0.2764	0.0147	0.5327	0.2749	-0.2578	0.516	1.0479
---	---	---	---	---	---	---	---	---	---
100	0.6995	0.6194	0.2301	0.0243	0.2185	0.2178	-0.4015	0.3517	1.1295

Hyperparameters:

Rank 1: {'reg__n_estimators': 700, 'reg__max_depth': 8, 'reg__learning_rate': 0.023369906439206178, 'reg__num_leaves': 25, 'reg__subsample': 0.7893987174371417, 'reg__colsample_bytree': 0.8129241396080022, 'reg__min_child_samples': 11, 'reg__reg_alpha': 6.551536543188579, 'reg__reg_lambda': 30.224381267681128, 'reg__min_split_gain': 1.5425747381371988, 'reg__path_smooth': 1.8100012584287204, 'reg__min_child_weight': 0.1022259369881337, 'reg__feature_fraction': 0.6479147548113109, 'reg__bagging_fraction': 0.639424429044574, 'reg__bagging_freq': 6, 'reg__max_bin': 159, 'reg__min_data_per_group': 69}

Rank 2: {'reg__n_estimators': 700, 'reg__max_depth': 6, 'reg__learning_rate': 0.022832567828820562, 'reg__num_leaves': 22, 'reg__subsample': 0.7997851343395538, 'reg__colsample_bytree': 0.8072558149969418, 'reg__min_child_samples': 46, 'reg__reg_alpha': 4.797291729069711, 'reg__reg_lambda': 14.537435207312752, 'reg__min_split_gain': 2.2648399983064826, 'reg__path_smooth': 4.99402278494587, 'reg__min_child_weight': 0.08979423129057636, 'reg__feature_fraction': 0.6484738599247394, 'reg__bagging_fraction': 0.729359777432482, 'reg__bagging_freq': 7, 'reg__max_bin': 146, 'reg__min_data_per_group': 66}

Rank 3: {'reg__n_estimators': 700, 'reg__max_depth': 6, 'reg__learning_rate': 0.02153962485834077, 'reg__num_leaves': 22, 'reg__subsample': 0.7164202919648174, 'reg__colsample_bytree': 0.8414785402785614, 'reg__min_child_samples': 45, 'reg__reg_alpha': 4.457718109335772, 'reg__reg_lambda': 32.88061354868932, 'reg__min_split_gain': 2.166541561994698, 'reg__path_smooth': 4.963269662158567, 'reg__min_child_weight': 0.07948062973404464, 'reg__feature_fraction': 0.6446299303550508, 'reg__bagging_fraction': 0.7284160001003014, 'reg__bagging_freq': 7, 'reg__max_bin': 146, 'reg__min_data_per_group': 67}

Rank 4: {'reg__n_estimators': 700, 'reg__max_depth': 8, 'reg__learning_rate': 0.02288904534727924, 'reg__num_leaves': 18, 'reg__subsample': 0.6768761679529076, 'reg__colsample_bytree': 0.7738039955668355, 'reg__min_child_samples': 16, 'reg__reg_alpha': 6.9694028272896364, 'reg__reg_lambda': 30.20211345098257, 'reg__min_split_gain': 1.9520019241983322, 'reg__path_smooth': 1.8933673897213652, 'reg__min_child_weight': 0.09848564754172093, 'reg__feature_fraction': 0.6139278325894906, 'reg__bagging_fraction': 0.7393370767815253, 'reg__bagging_freq': 6, 'reg__max_bin': 176, 'reg__min_data_per_group': 72}

Rank 5: {'reg__n_estimators': 600, 'reg__max_depth': 7, 'reg__learning_rate': 0.016065610578107358, 'reg__num_leaves': 37, 'reg__subsample': 0.8418421884324485, 'reg__colsample_bytree': 0.6794914665830591, 'reg__min_child_samples': 13, 'reg__reg_alpha': 5.776645116299782, 'reg__reg_lambda': 11.554188373914466, 'reg__min_split_gain': 2.505485230569258, 'reg__path_smooth': 0.5733518784038771, 'reg__min_child_weight': 0.1318897554077529, 'reg__feature_fraction': 0.6059942176712435, 'reg__bagging_fraction': 0.6529760980461063, 'reg__bagging_freq': 5, 'reg__max_bin': 167, 'reg__min_data_per_group': 75}

Rank 6: {'reg__n_estimators': 800, 'reg__max_depth': 4, 'reg__learning_rate': 0.09808051366250824, 'reg__num_leaves': 19, 'reg__subsample': 0.8250661135431144, 'reg__colsample_bytree': 0.5958574567796663, 'reg__min_child_samples': 13, 'reg__reg_alpha': 7.26991046837138, 'reg__reg_lambda': 26.547672417892514, 'reg__min_split_gain': 1.5546578415303824, 'reg__path_smooth': 4.9346042978366205, 'reg__min_child_weight': 0.04254565165440046, 'reg__feature_fraction': 0.7999583986971248, 'reg__bagging_fraction': 0.6502012080615758, 'reg__bagging_freq': 7, 'reg__max_bin': 180, 'reg__min_data_per_group': 60}

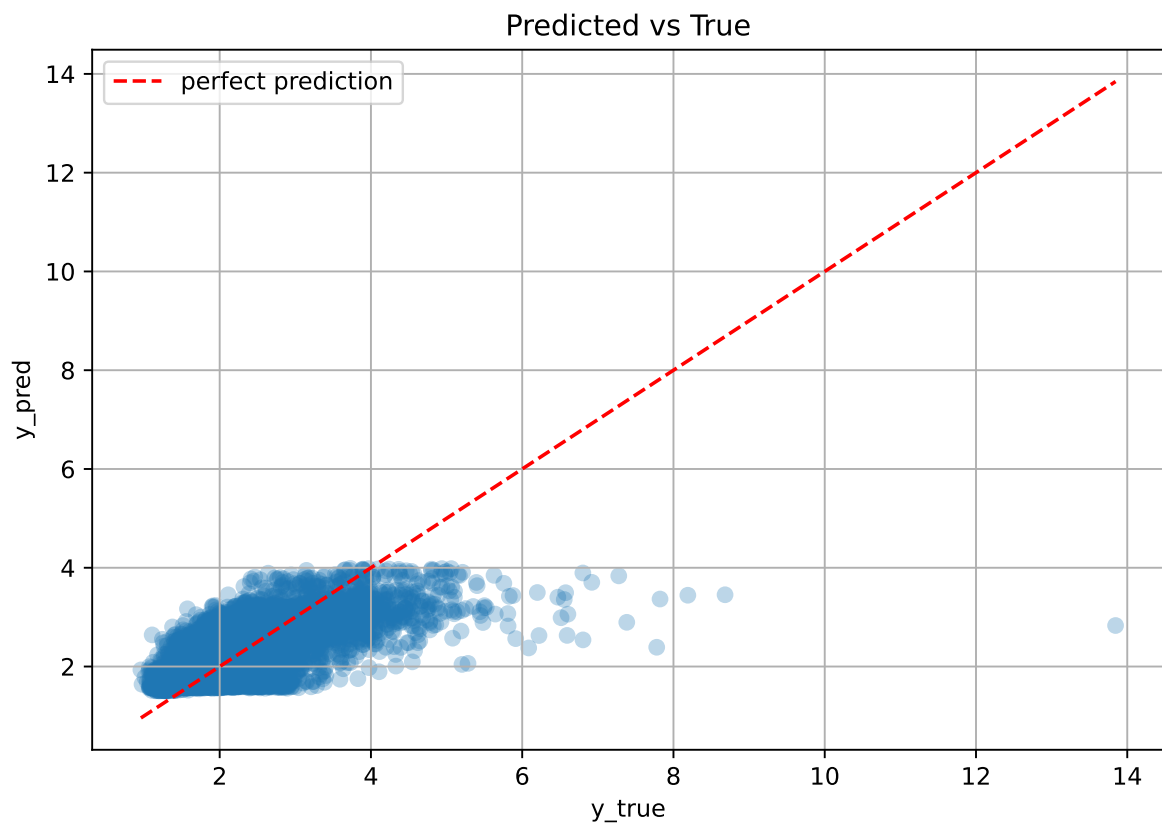
Rank 7: {'reg__n_estimators': 800, 'reg__max_depth': 7, 'reg__learning_rate': 0.010626043171579787, 'reg__num_leaves': 25, 'reg__subsample': 0.891357054219115, 'reg__colsample_bytree': 0.6111392521283086, 'reg__min_child_samples': 22, 'reg__reg_alpha': 8.50039928809161, 'reg__reg_lambda': 17.211643059585203, 'reg__min_split_gain': 1.7540346015893524, 'reg__path_smooth': 1.6074635159069417, 'reg__min_child_weight': 0.011263276017585628, 'reg__feature_fraction': 0.6728088486798753, 'reg__bagging_fraction': 0.6313457975093484, 'reg__bagging_freq': 6, 'reg__max_bin': 119, 'reg__min_data_per_group': 94}

Rank 8: {'reg__n_estimators': 700, 'reg__max_depth': 8, 'reg__learning_rate': 0.020562075697922625, 'reg__num_leaves': 34, 'reg__subsample': 0.7872169532684097, 'reg__colsample_bytree': 0.8892106620956592, 'reg__min_child_samples': 14, 'reg__reg_alpha': 6.417455052507133, 'reg__reg_lambda': 38.40114945958329, 'reg__min_split_gain': 2.089179450039656, 'reg__path_smooth': 2.157112647225892, 'reg__min_child_weight': 0.054080122957527565, 'reg__feature_fraction': 0.6143434275245004, 'reg__bagging_fraction': 0.7034124939717425, 'reg__bagging_freq': 5, 'reg__max_bin': 174, 'reg__min_data_per_group': 81}

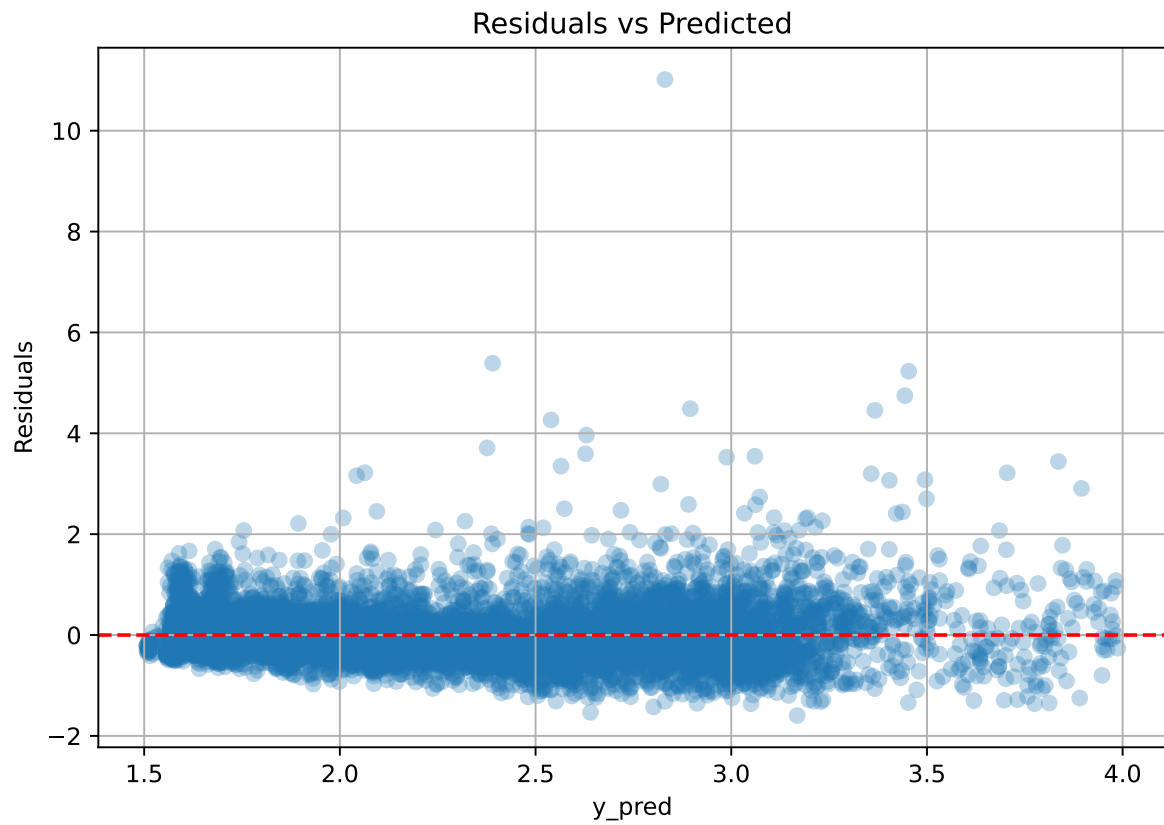
Rank 9: {'reg__n_estimators': 700, 'reg__max_depth': 8, 'reg__learning_rate': 0.023567434487563505, 'reg__num_leaves': 32, 'reg__subsample': 0.807263240896618, 'reg__colsample_bytree': 0.6339802336047335, 'reg__min_child_samples': 19, 'reg__reg_alpha': 7.842909174944697, 'reg__reg_lambda': 13.069343481347994, 'reg__min_split_gain': 2.322473944829824, 'reg__path_smooth': 1.1862931807753094, 'reg__min_child_weight': 0.07014149498058027, 'reg__feature_fraction': 0.646551103239295, 'reg__bagging_fraction': 0.7257714831789632, 'reg__bagging_freq': 7, 'reg__max_bin': 187, 'reg__min_data_per_group': 70}

Rank 10: {'reg__n_estimators': 800, 'reg__max_depth': 5, 'reg__learning_rate': 0.01154377685909759, 'reg__num_leaves': 16, 'reg__subsample': 0.7637256771950205, 'reg__colsample_bytree': 0.8944508425070017, 'reg__min_child_samples': 42, 'reg__reg_alpha': 7.236282352269628, 'reg__reg_lambda': 49.83126323026745, 'reg__min_split_gain': 1.995550258985188, 'reg__path_smooth': 1.4075006537074932, 'reg__min_child_weight': 0.013155932269246872, 'reg__feature_fraction': 0.6776420826194612, 'reg__bagging_fraction': 0.773434521539058, 'reg__bagging_freq': 6, 'reg__max_bin': 212, 'reg__min_data_per_group': 116}

2.4.2 Scatter



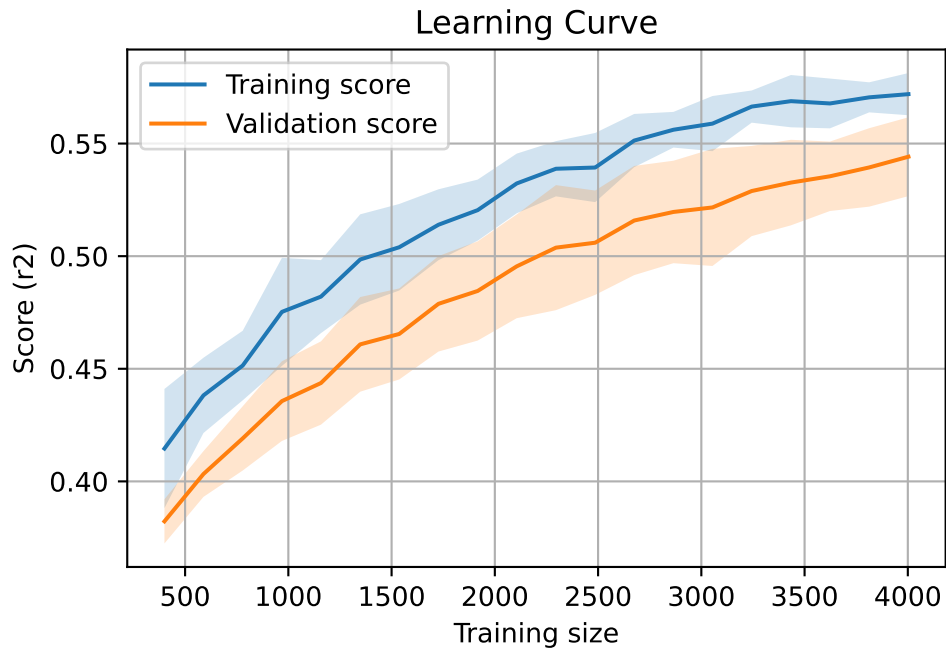
2.4.3 Scatter residuals



2.4.4 Learning curve

Len(group) : 8168, **Len(training)** : 5718,

Len(training_real) : 4003, **Len(test_of_training)** : 1715, **Len(test)** : 2450

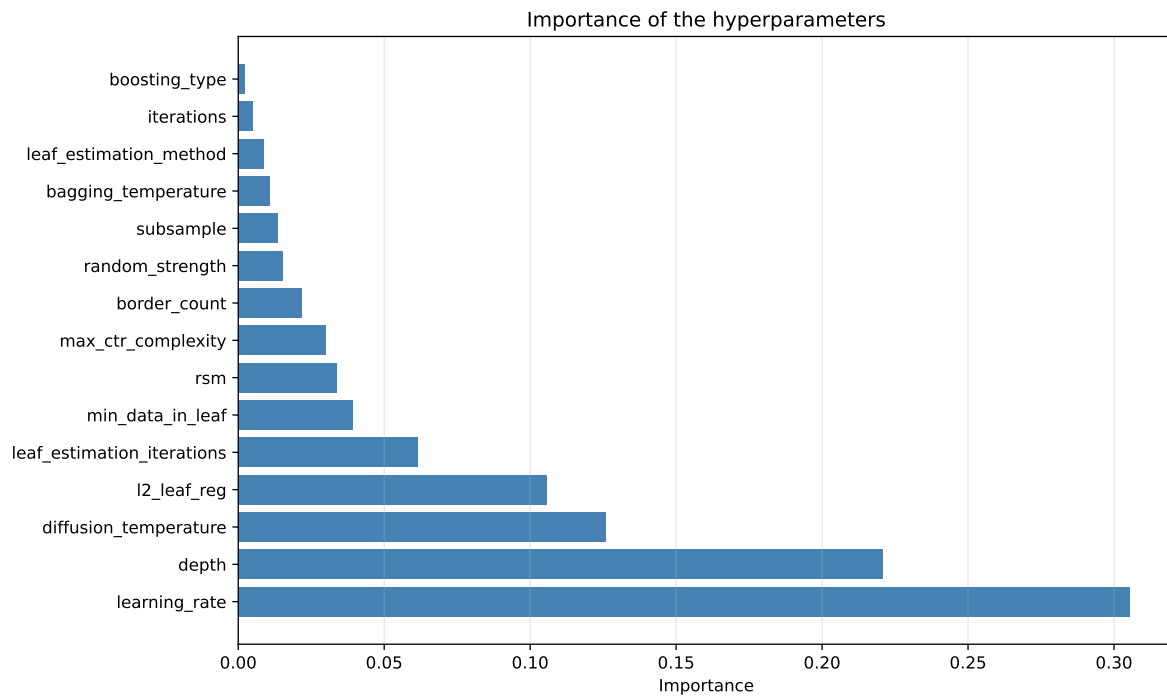
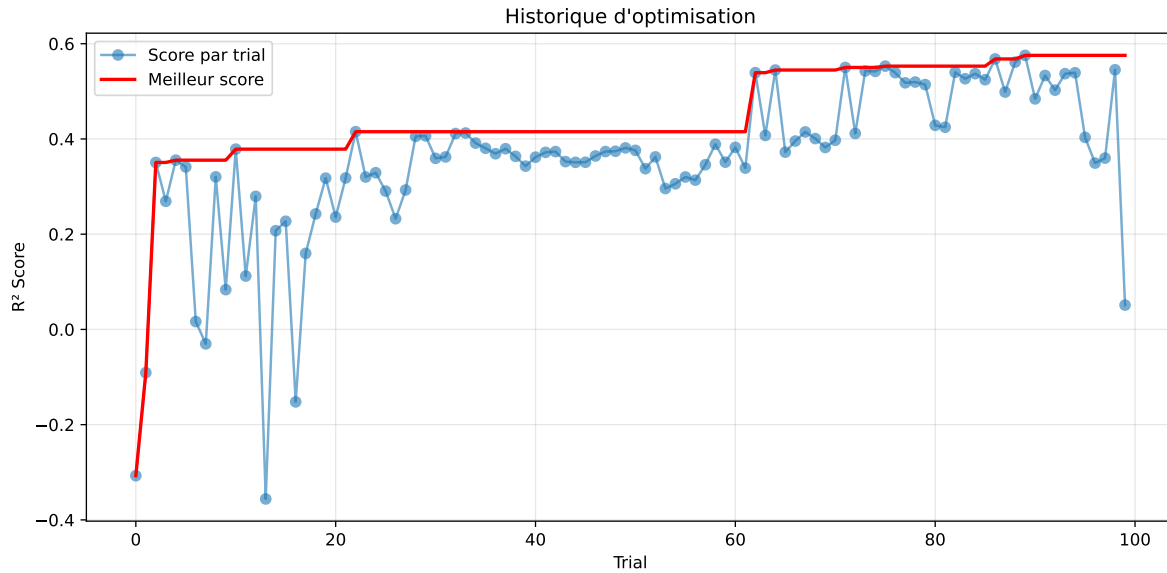


2.5 Model : CatBoost

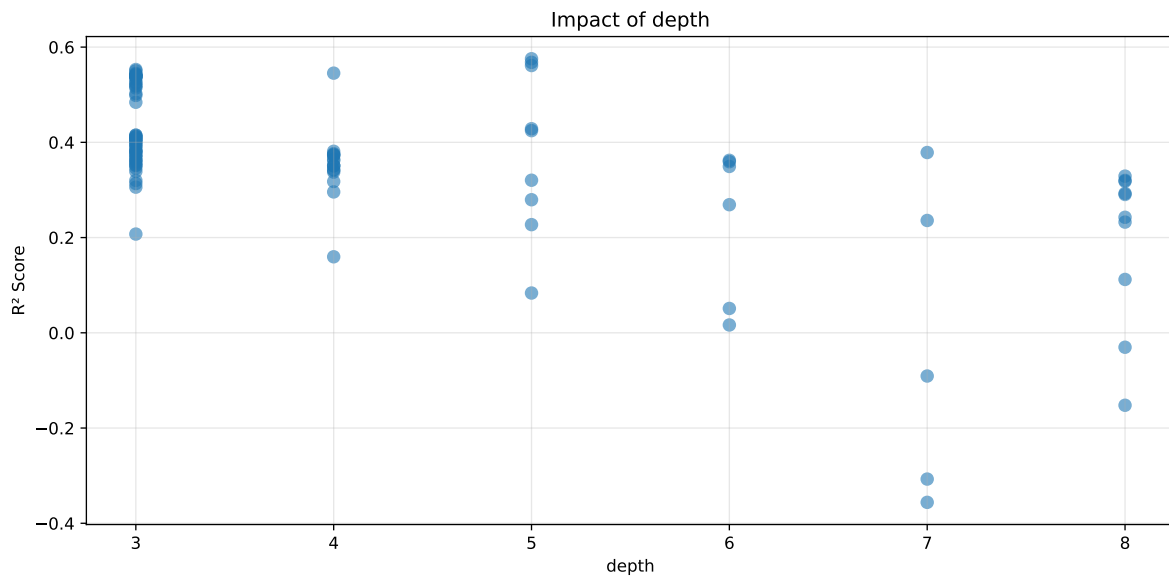
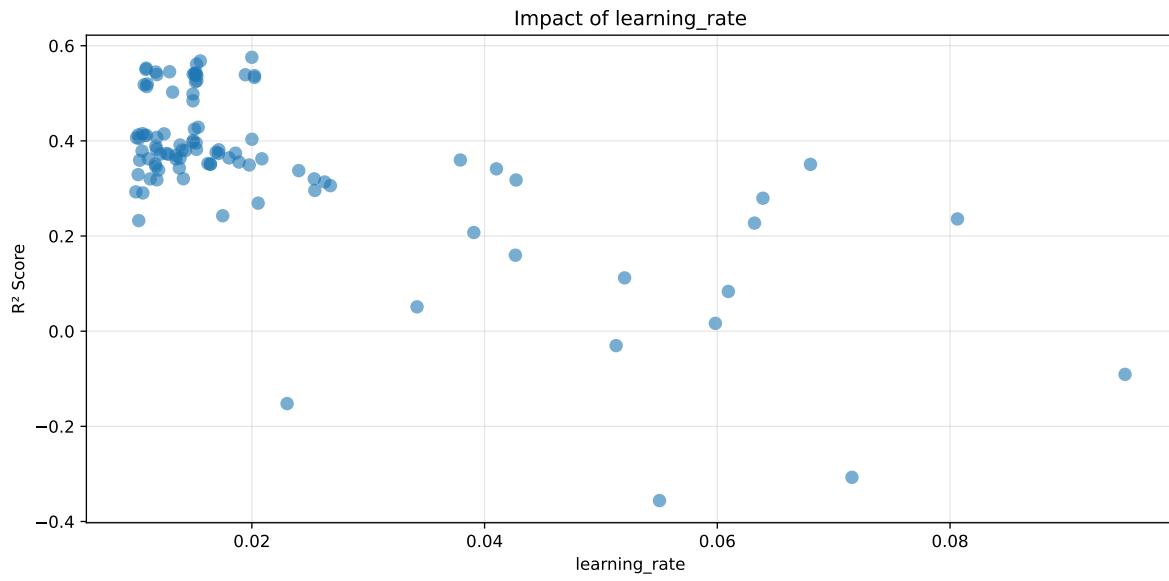
2.5.1 Gridsearch

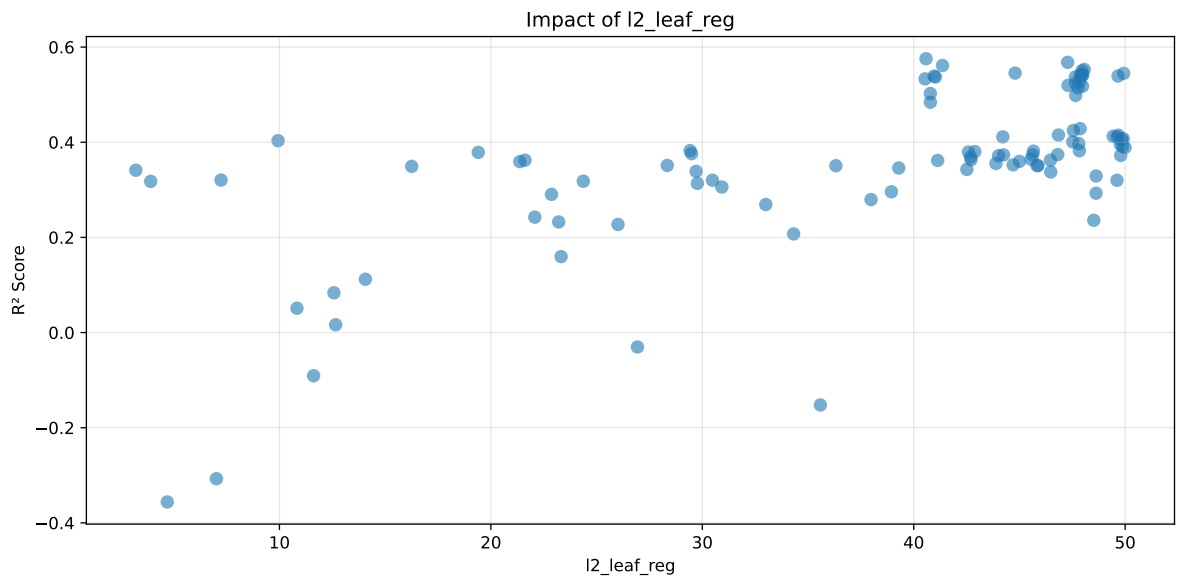
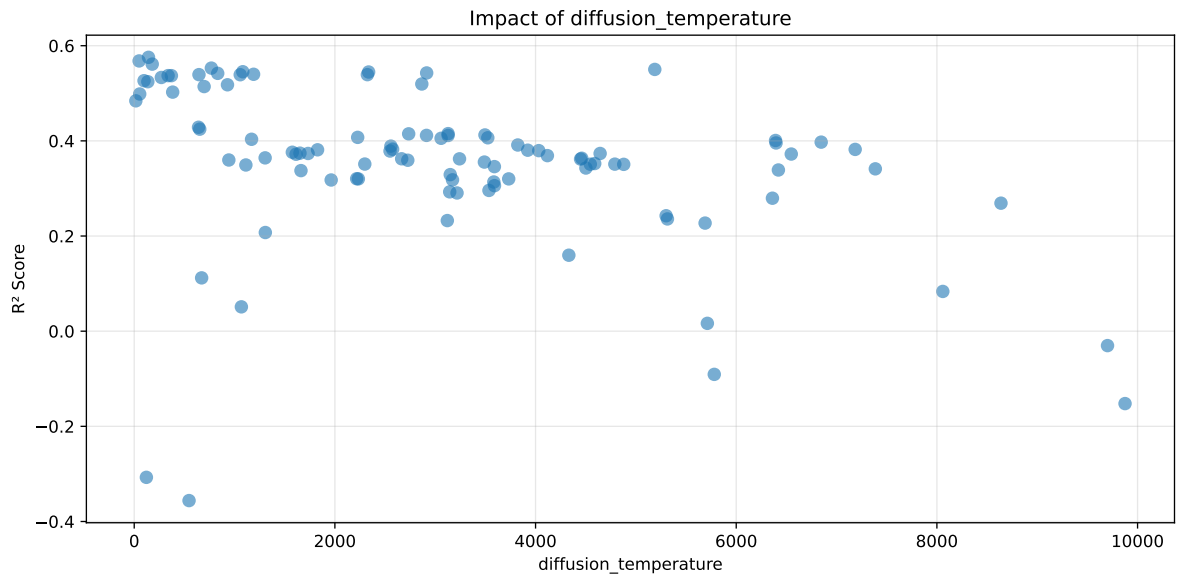
Distribution y_train vs y_test: y_train: mean=2.354, std=0.804, min=0.959, max=8.682
y_test: mean=2.340, std=0.772, min=1.071, max=13.845

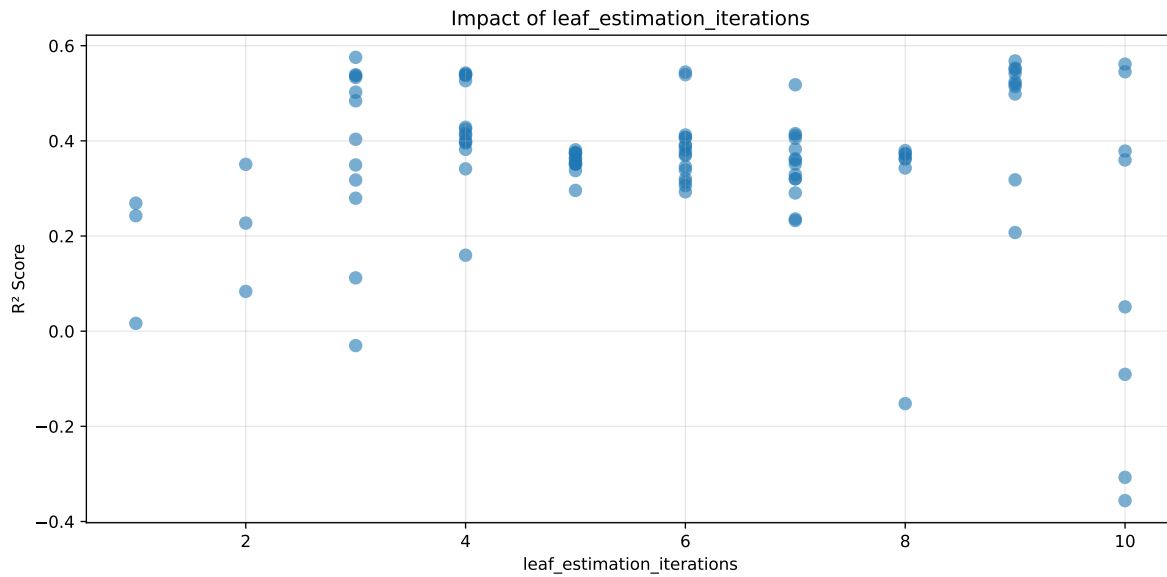
===== TOP Importance FEATURES =====
feature importance r_umidity 41.567611 r_CO2 18.848223 r_NH3 16.630745 r_THI 12.810522
r_temperature 10.142898



Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'diffusion_temperature', 'l2_leaf_reg', 'leaf_estimation_iterations']

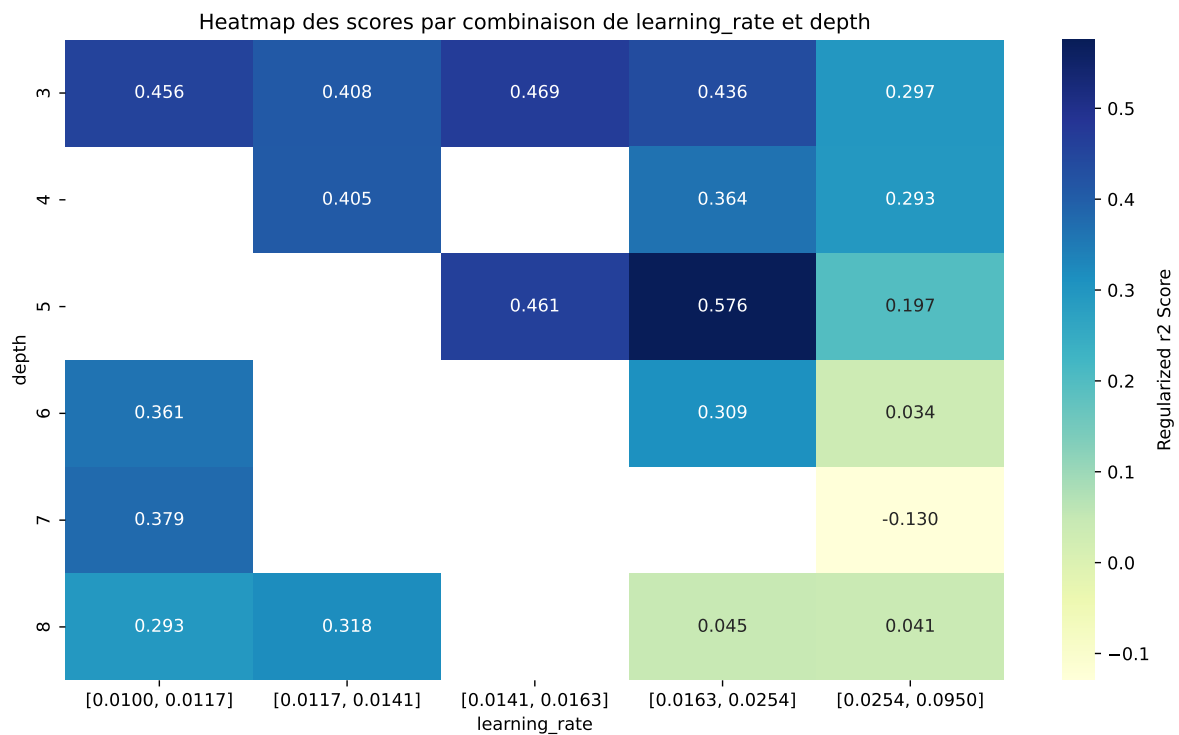






C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6024	0.5756	0.2902	0.0188	0.5756	0.2878	-0.2878	0.5	1.0465
2	0.5932	0.568	0.2873	0.0176	0.568	0.2857	-0.2823	0.5029	1.0444
3	0.5857	0.5613	0.2923	0.0176	0.5613	0.2872	-0.2741	0.5116	1.0435
4	0.577	0.5529	0.2884	0.0185	0.5529	0.283	-0.2699	0.5118	1.0436
5	0.574	0.5502	0.2823	0.0197	0.5502	0.2872	-0.2629	0.5221	1.0434
6	0.5684	0.5453	0.2948	0.0172	0.5453	0.2902	-0.2551	0.5322	1.0423
7	0.5686	0.5448	0.291	0.0178	0.5448	0.2865	-0.2582	0.526	1.0437
8	0.5649	0.5427	0.2902	0.0175	0.5427	0.2857	-0.257	0.5264	1.0408
9	0.5635	0.5419	0.2902	0.0184	0.5419	0.2851	-0.2569	0.526	1.0398
10	0.5608	0.54	0.2871	0.0173	0.54	0.2844	-0.2556	0.5266	1.0385
---	---	---	---	---	---	---	---	---	---
100	0.8615	0.6586	0.2068	0.0216	-0.3561	0.2147	-0.4439	0.3261	1.3081

Hyperparameters:

Rank 1: {'reg__iterations': 400, 'reg__depth': 5, 'reg__learning_rate': 0.019997225605122806, 'reg__l2_leaf_reg': 40.580734428948496, 'reg__bagging_temperature': 0.9574711189950448, 'reg__random_strength': 2.3645842253950775, 'reg__subsample': 0.8877297352953925, 'reg__min_data_in_leaf': 49, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.6887416026989277, 'reg__border_count': 94, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 142.37252664225014}

Rank 2: {'reg__iterations': 500, 'reg__depth': 5, 'reg__learning_rate': 0.01556996791371721, 'reg__l2_leaf_reg': 47.2800250936373, 'reg__bagging_temperature': 1.445243855376867, 'reg__random_strength': 2.0998188698812084, 'reg__subsample': 0.5449296371711089, 'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.6970261077956857, 'reg__border_count': 98, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 48.96229921869576}

Rank 3: {'reg__iterations': 400, 'reg__depth': 5, 'reg__learning_rate': 0.015242198062885745, 'reg__l2_leaf_reg': 41.36030240201737, 'reg__bagging_temperature': 2.456227071335976, 'reg__random_strength': 2.3751385098310034, 'reg__subsample': 0.8972439305321815, 'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.7607646518327028, 'reg__border_count': 92, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 179.52657395439877}

Rank 4: {'reg__iterations': 800, 'reg__depth': 3, 'reg__learning_rate': 0.010923791557643517, 'reg__l2_leaf_reg': 48.06427845373739, 'reg__bagging_temperature': 1.429657206552244, 'reg__random_strength': 2.1402457337623235, 'reg__subsample': 0.8120028376929045, 'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 9, 'reg__rsm':

0.8969576886363361, 'reg__border_count': 99, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 768.7991752628427}

Rank 5: {'reg__iterations': 800, 'reg__depth': 3, 'reg__learning_rate': 0.010896896883926357, 'reg__l2_leaf_reg': 47.95620008445596, 'reg__bagging_temperature': 0.7187741252826514, 'reg__random_strength': 2.240337230673056, 'reg__subsample': 0.8248246935411268, 'reg__min_data_in_leaf': 38, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.8930619827900068, 'reg__border_count': 103, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 5188.374963110076}

Rank 6: {'reg__iterations': 400, 'reg__depth': 4, 'reg__learning_rate': 0.012928722771262979, 'reg__l2_leaf_reg': 44.79225571128832, 'reg__bagging_temperature': 1.7706672311086928, 'reg__random_strength': 1.7932217845737737, 'reg__subsample': 0.8637310759323897, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.7599845200006348, 'reg__border_count': 91, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 1082.0443888391142}

Rank 7: {'reg__iterations': 600, 'reg__depth': 3, 'reg__learning_rate': 0.011741749360880275, 'reg__l2_leaf_reg': 49.93050555187259, 'reg__bagging_temperature': 1.8013983063297392, 'reg__random_strength': 1.5946750115559691, 'reg__subsample': 0.7818988254206759, 'reg__min_data_in_leaf': 41, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.8293576014807266, 'reg__border_count': 106, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 2336.1886573254287}

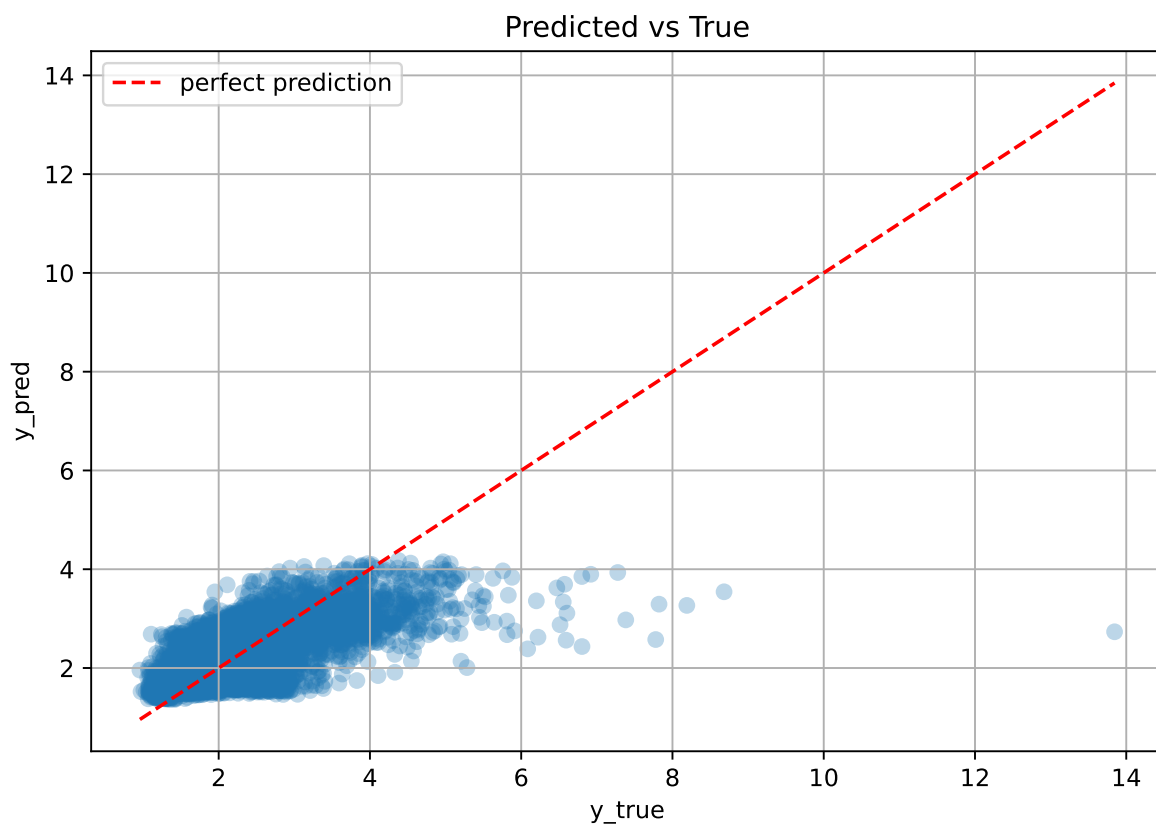
Rank 8: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.015212306541608198, 'reg__l2_leaf_reg': 47.90202770840751, 'reg__bagging_temperature': 0.7334051397381757, 'reg__random_strength': 2.185362244786154, 'reg__subsample': 0.8997132416384318, 'reg__min_data_in_leaf': 47, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.8986665231805744, 'reg__border_count': 105, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 2914.7295361349065}

Rank 9: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.01508028400158332, 'reg__l2_leaf_reg': 47.9951714520829, 'reg__bagging_temperature': 0.8167583243527368, 'reg__random_strength': 2.1838988006772526, 'reg__subsample': 0.5593242729791741, 'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.7922284983470993, 'reg__border_count': 102, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 832.1703054135469}

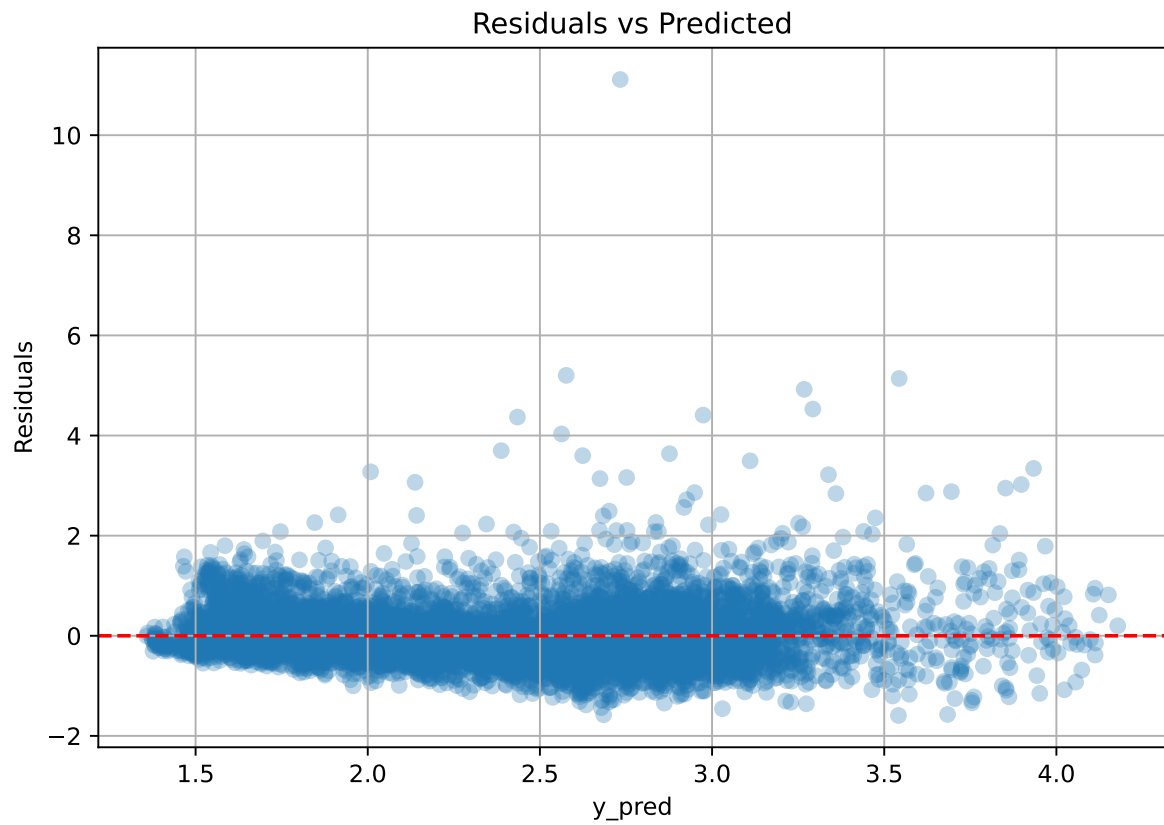
Rank 10: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.014956692997484738, 'reg__l2_leaf_reg': 47.9634074327297, 'reg__bagging_temperature': 1.4176894651090968,

```
'reg__random_strength': 2.087648056581105, 'reg__subsample': 0.8806640275417523,  
'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 4, 'reg__rsm':  
0.8083968916014549, 'reg__border_count': 90, 'reg__leaf_estimation_method': 'Gradient',  
'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature':  
1189.6482899476255}
```

2.5.2 Scatter



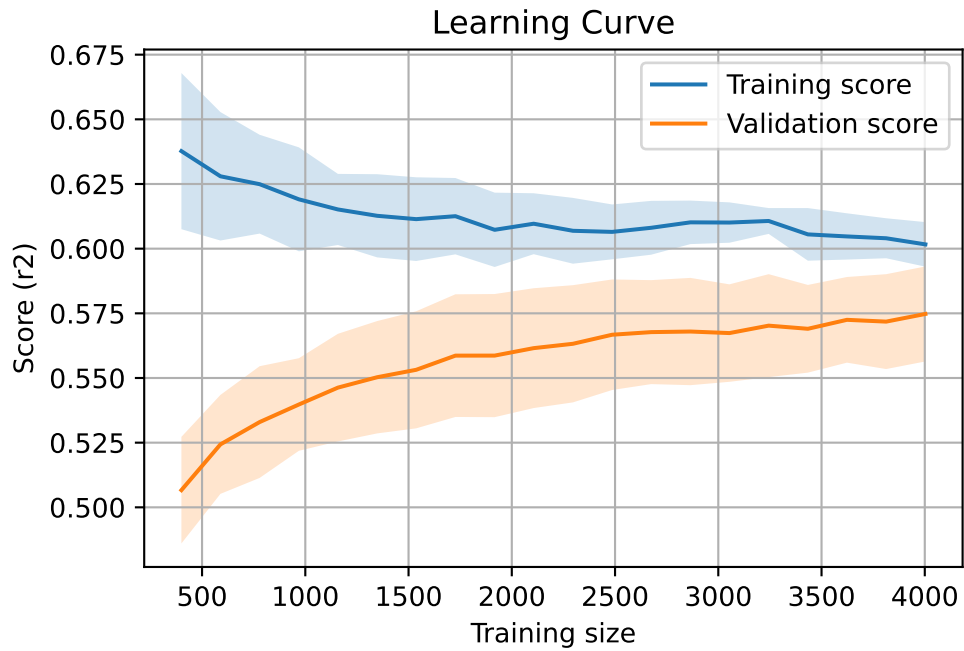
2.5.3 Scatter residuals



2.5.4 Learning curve

`Len(group)` : 8168, `Len(training)` : 5718,

`Len(training_real)` : 4003, `Len(test_of_training)` : 1715, `Len(test)` : 2450



2.6 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.403	0.408	0.011	0.187	0.187	-0.222	0.457	0.987	nan
LGBM	0.573	0.546	0.018	0.279	0.273	-0.273	0.5	1.049	0.5464
CatBoost	0.602	0.576	0.019	0.29	0.288	-0.288	0.5	1.046	0.5756

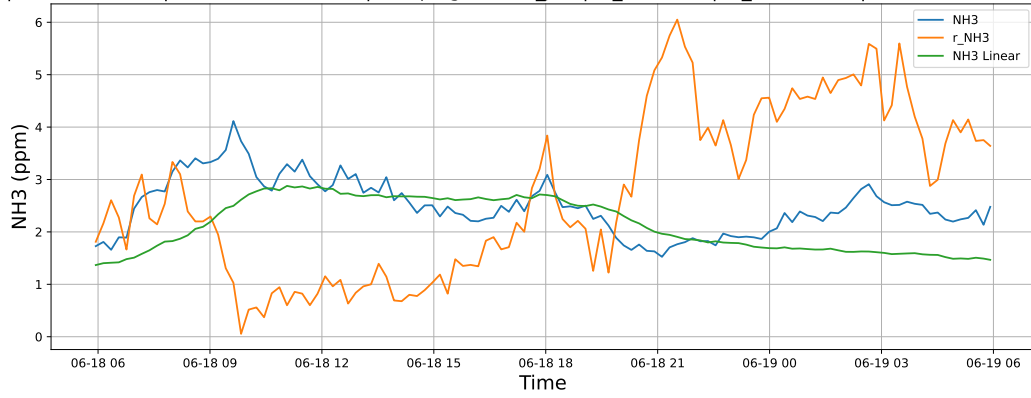
3 Plot

3.1 Standalone plots

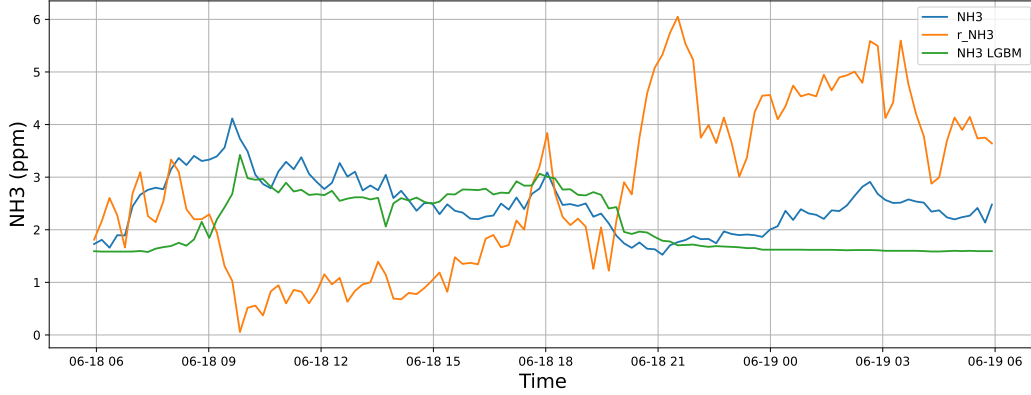
3.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.1.1.1 Time window : 24

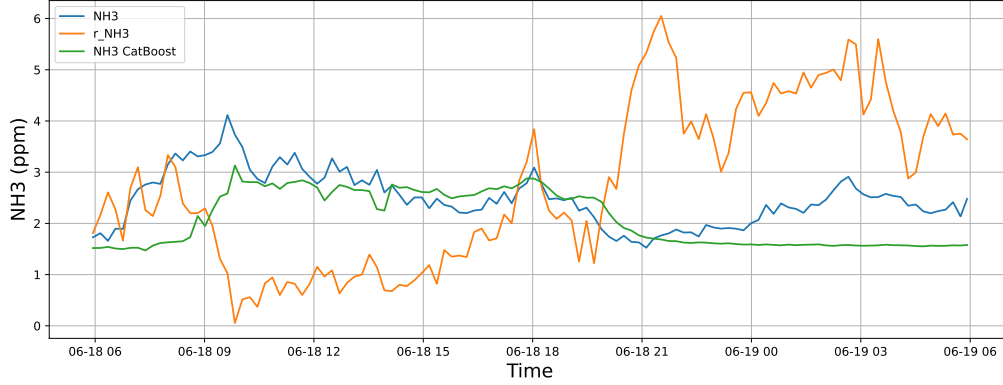
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

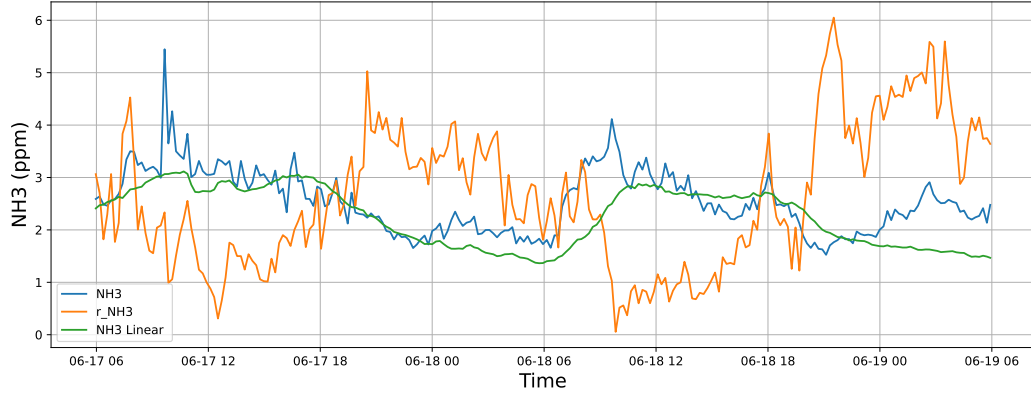


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

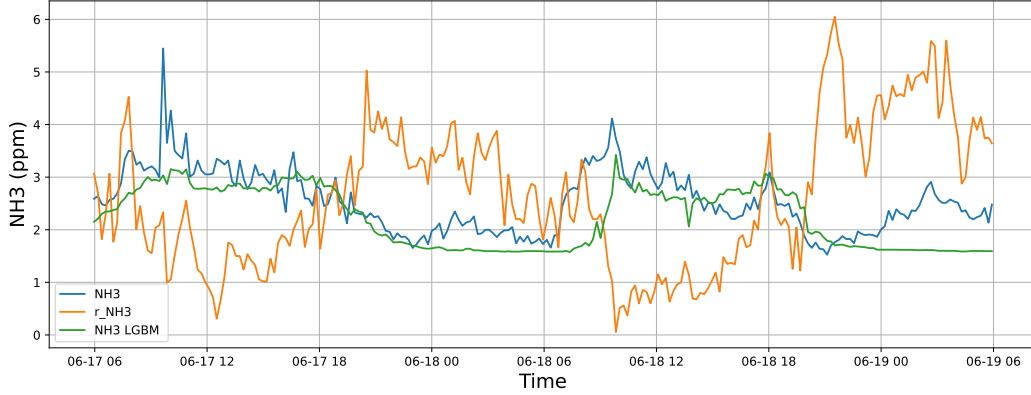


3.1.1.2 Time window : 48

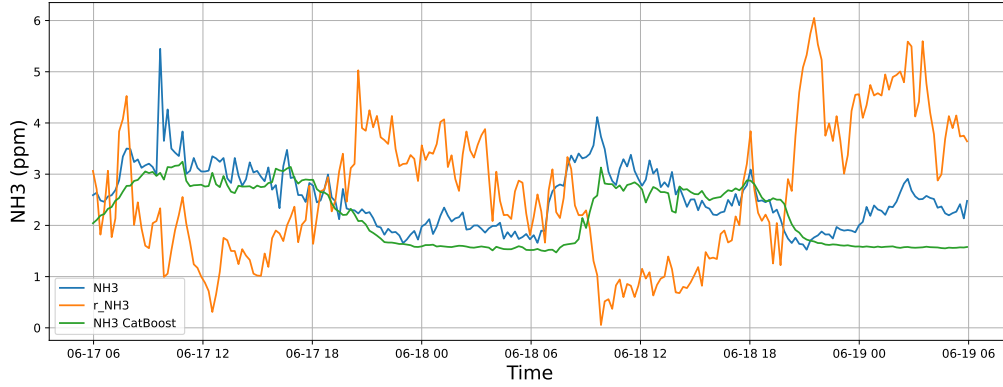
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

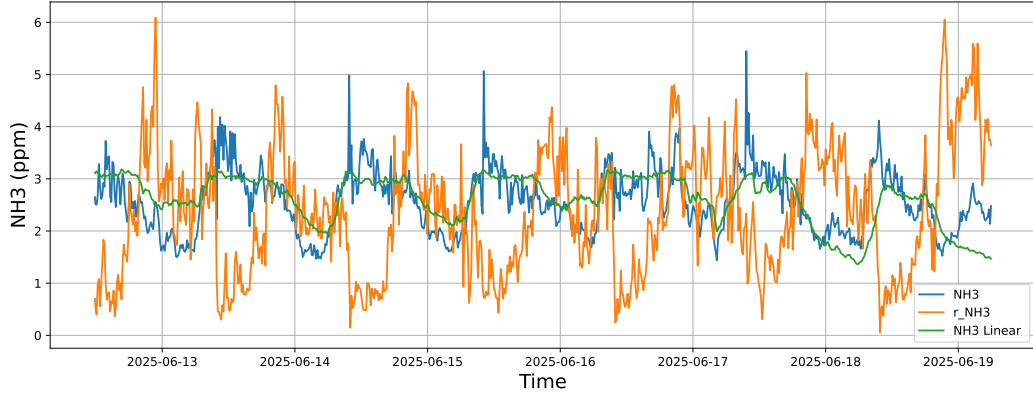


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

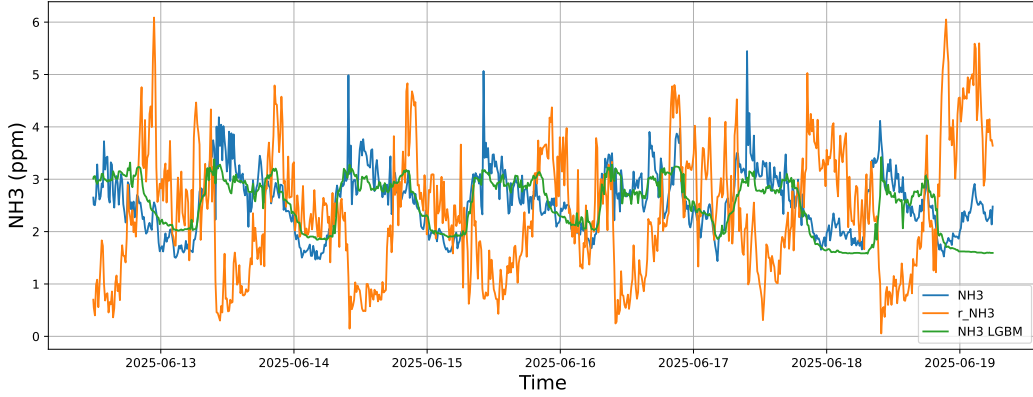


3.1.1.3 Time window : ALL

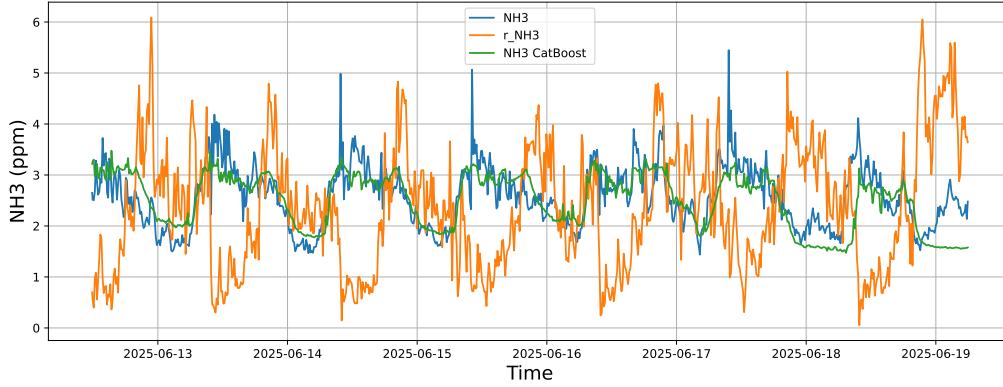
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



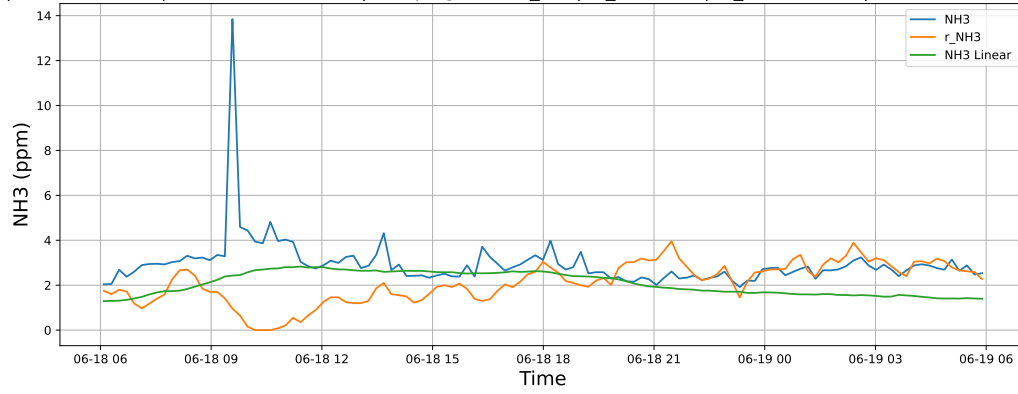
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



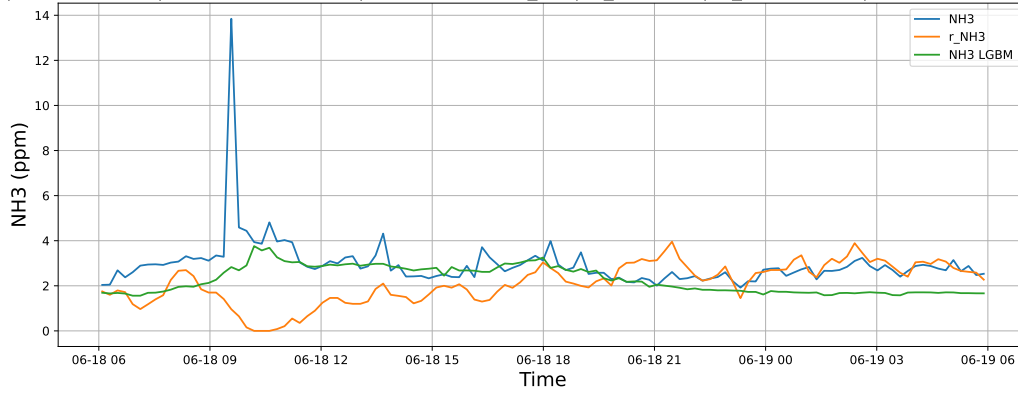
3.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

3.1.2.1 Time window : 24

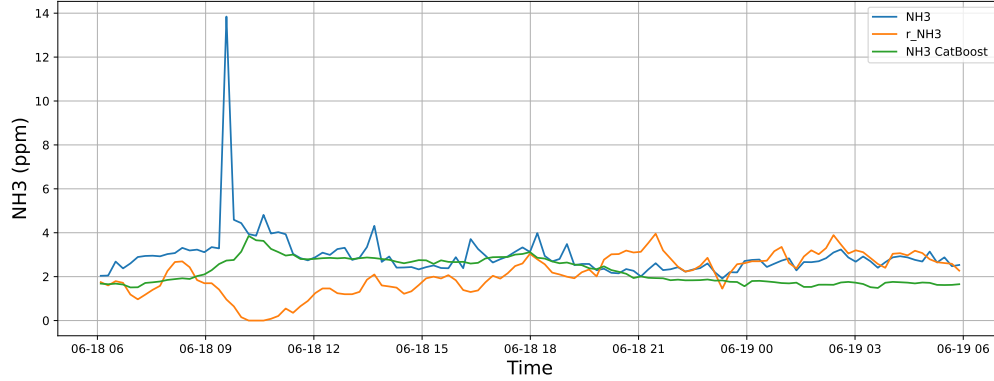
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

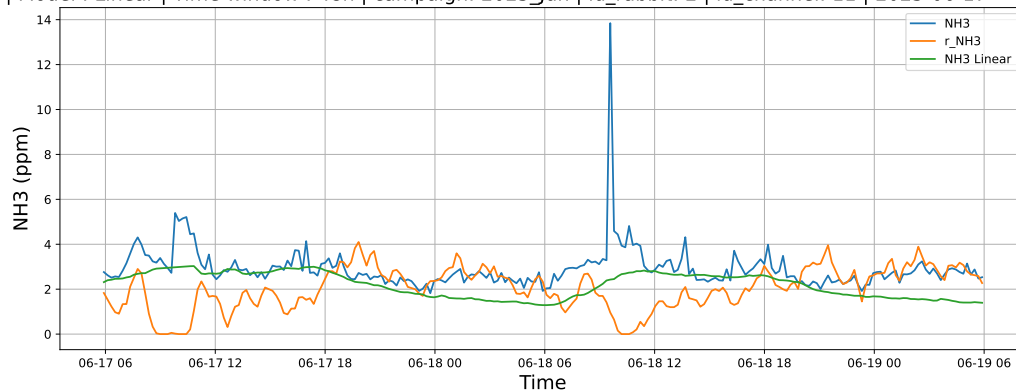


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

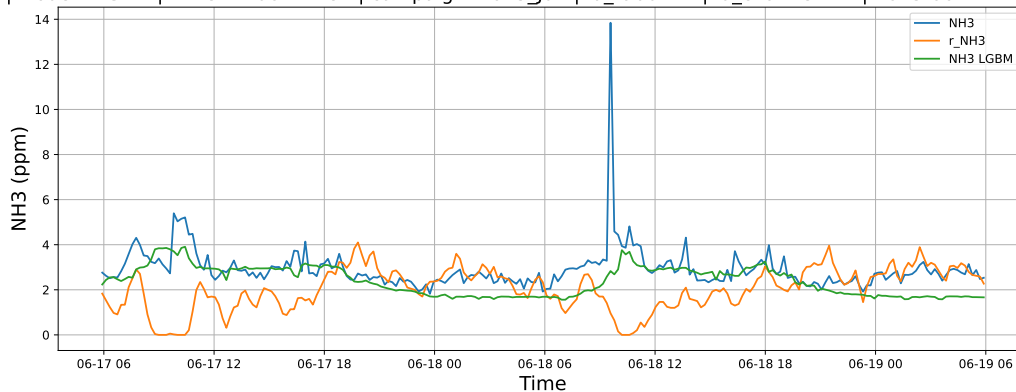


3.1.2.2 Time window : 48

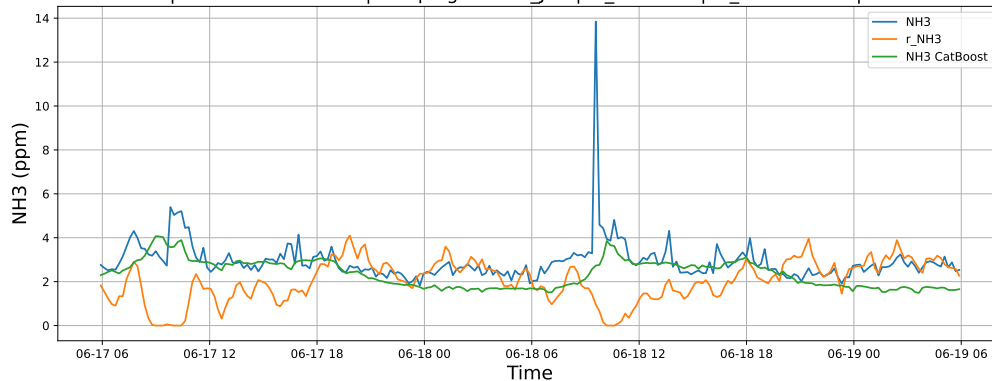
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

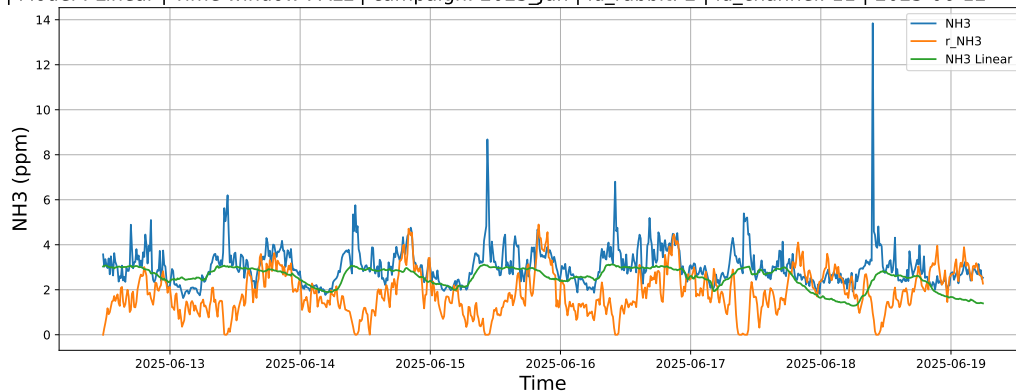


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

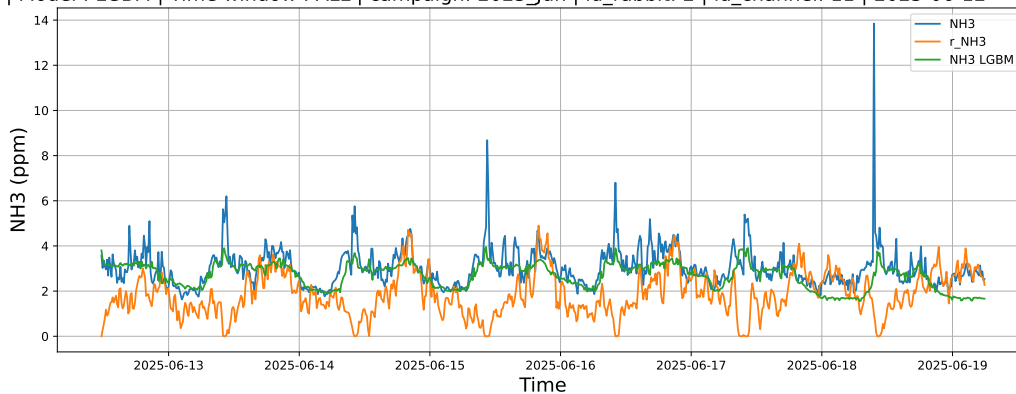


3.1.2.3 Time window : ALL

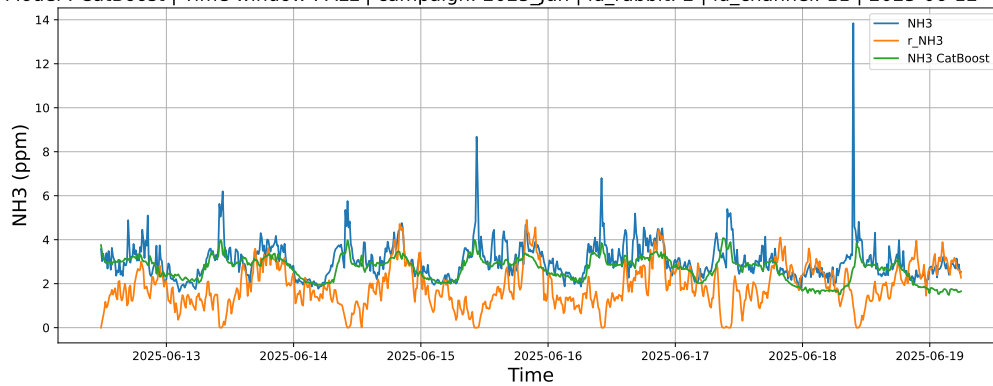
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



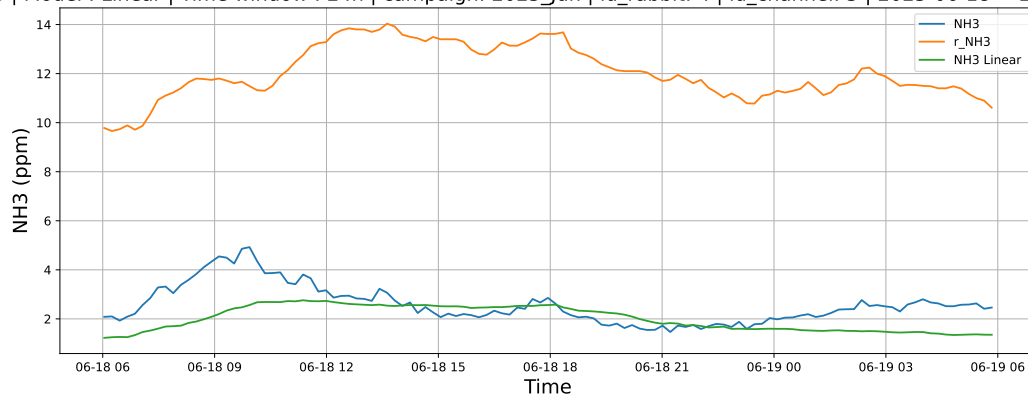
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



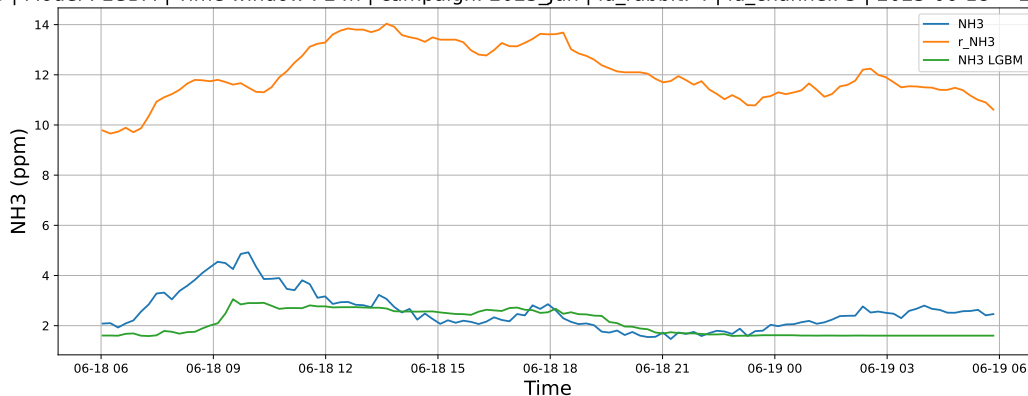
3.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

3.1.3.1 Time window : 24

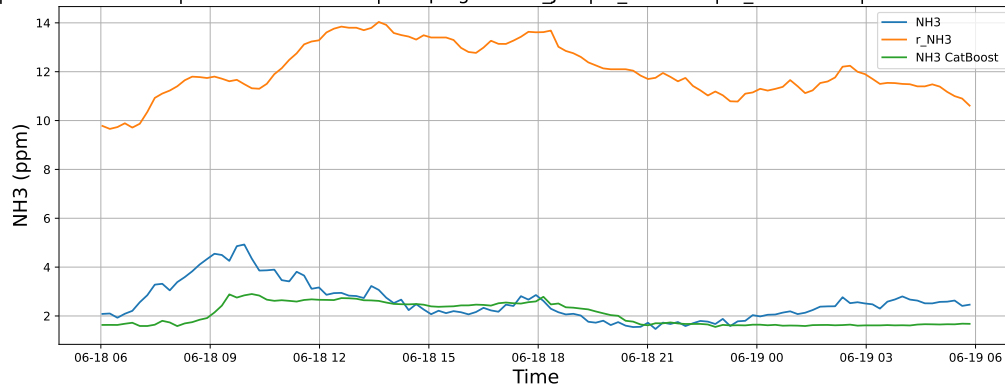
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

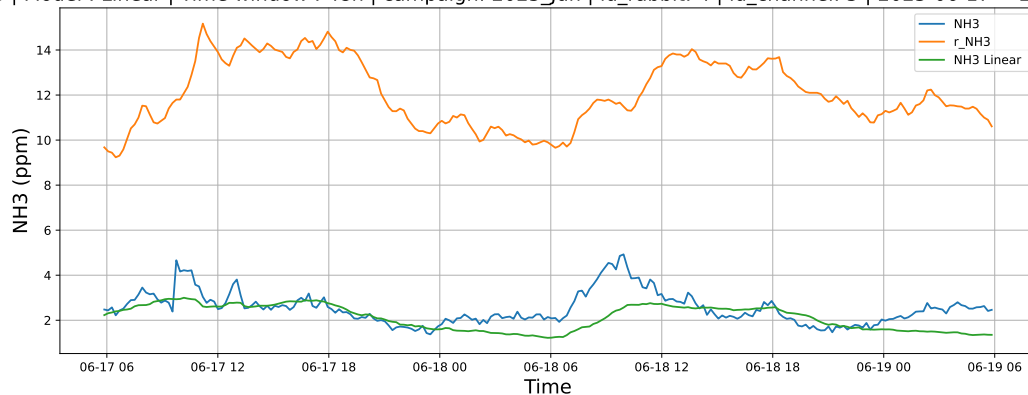


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

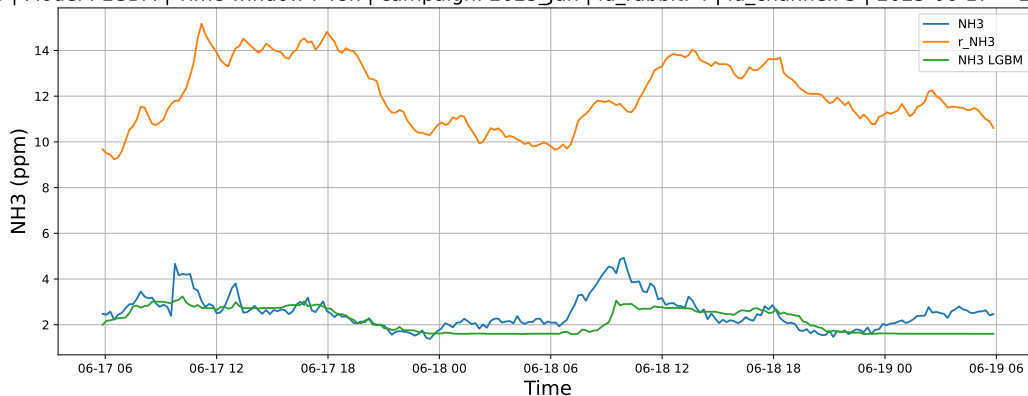


3.1.3.2 Time window : 48

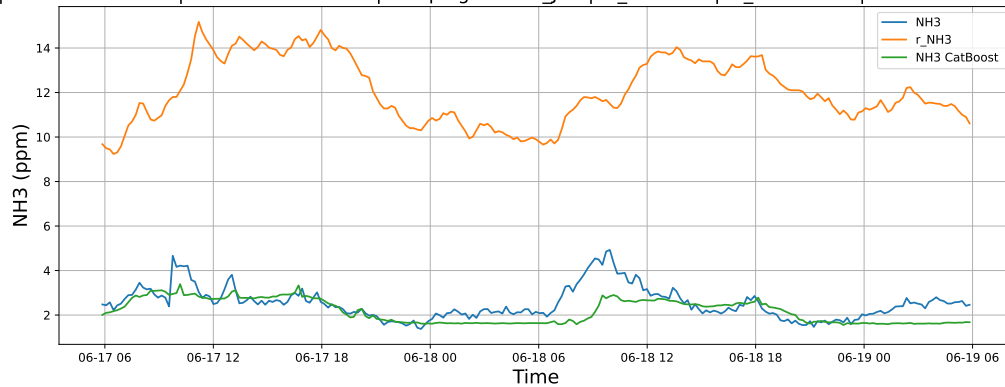
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

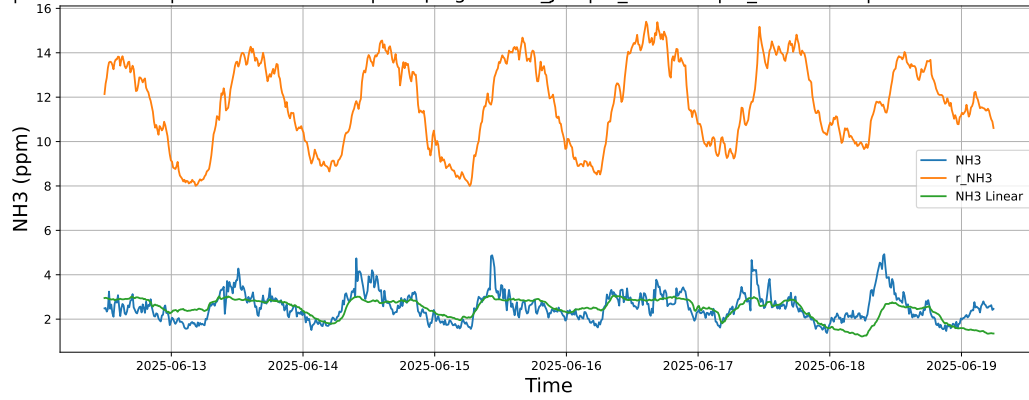


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

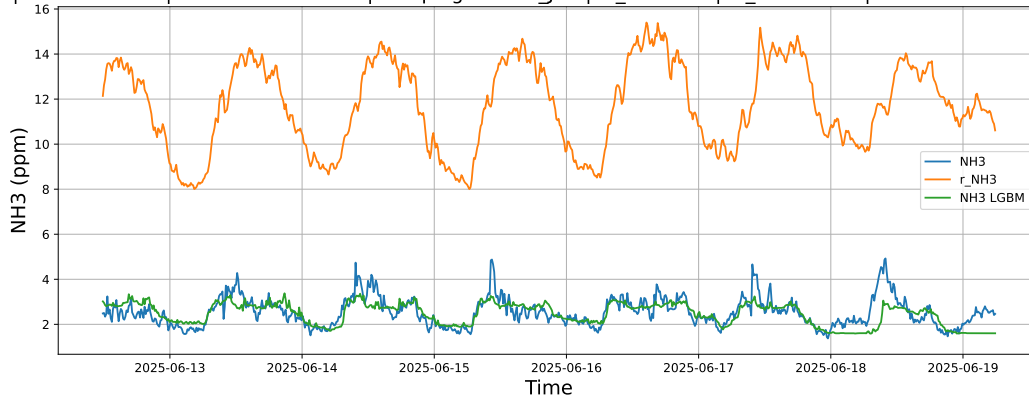


3.1.3.3 Time window : ALL

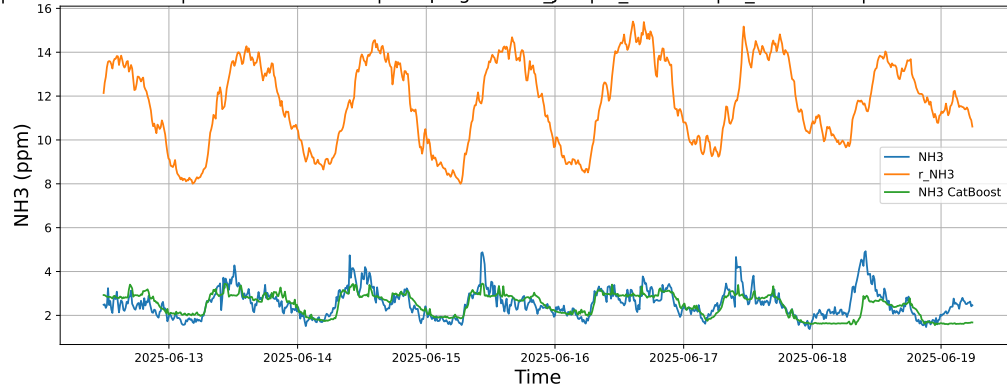
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



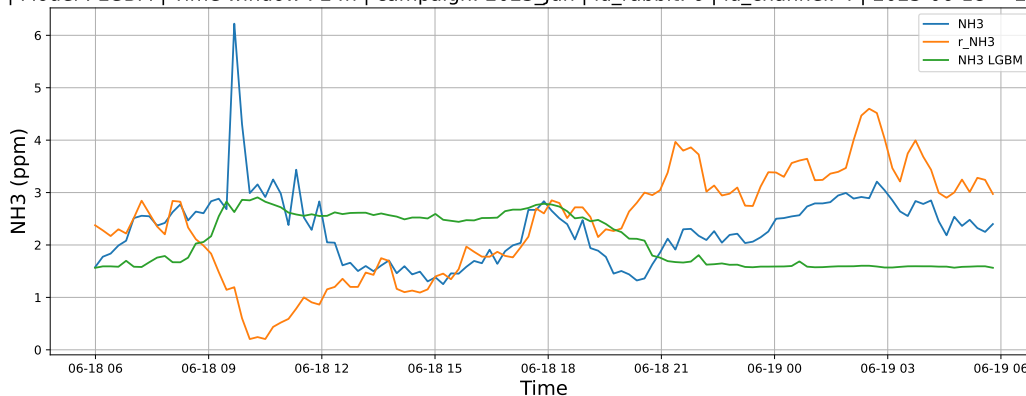
3.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

3.1.4.1 Time window : 24

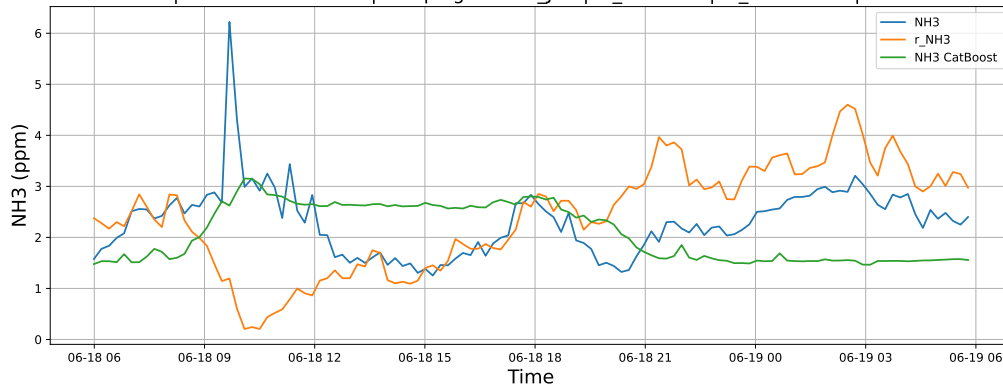
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

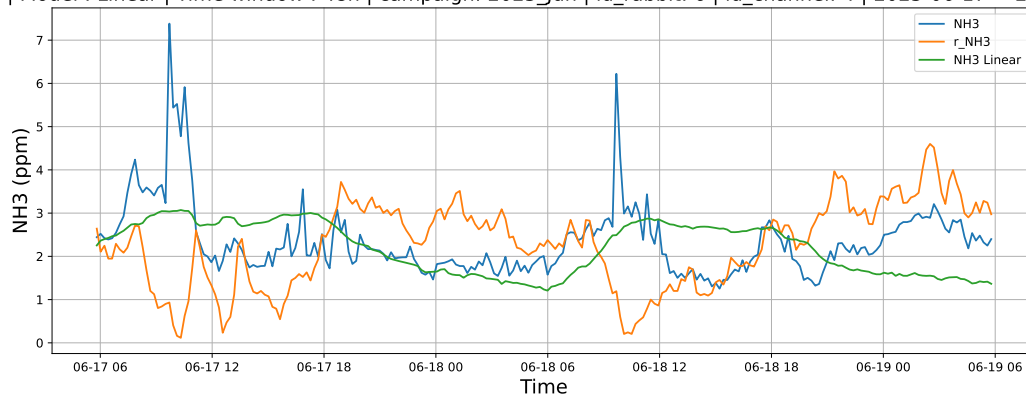


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

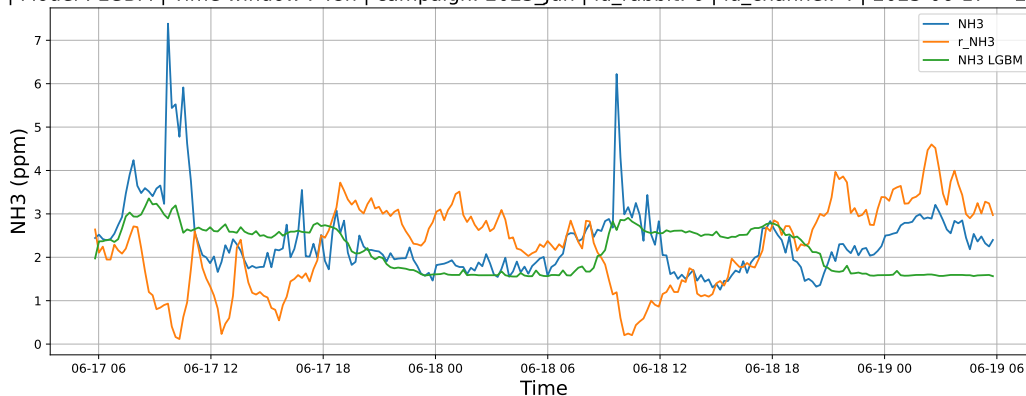


3.1.4.2 Time window : 48

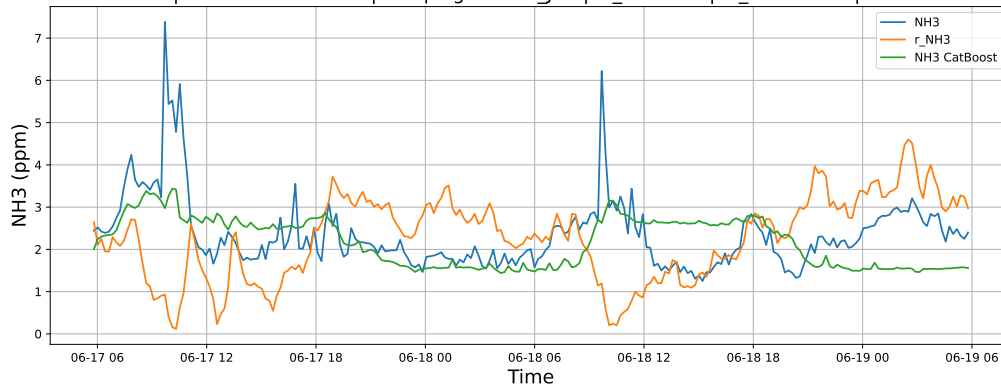
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

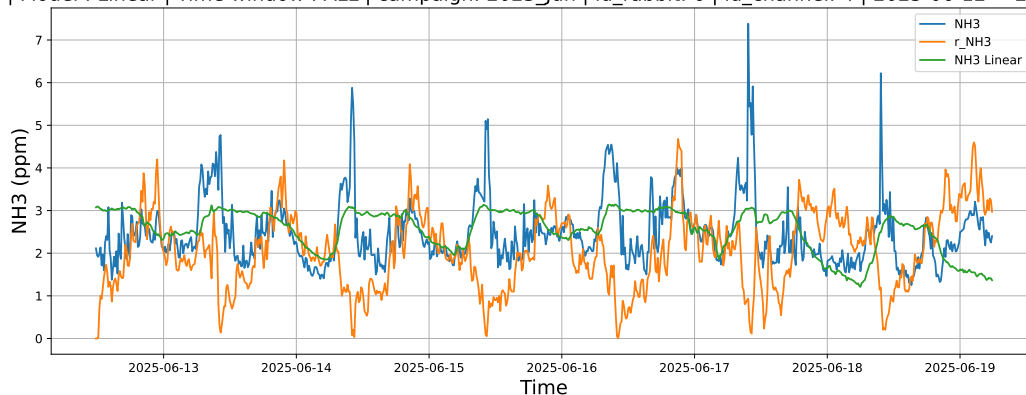


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

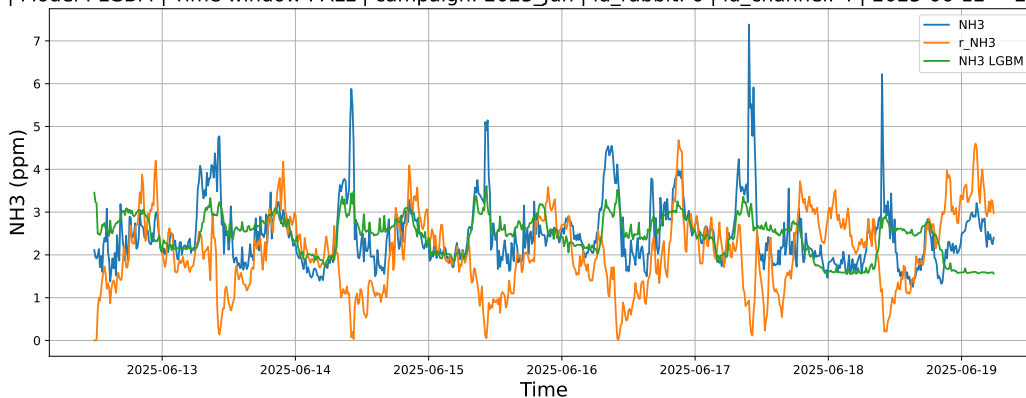


3.1.4.3 Time window : ALL

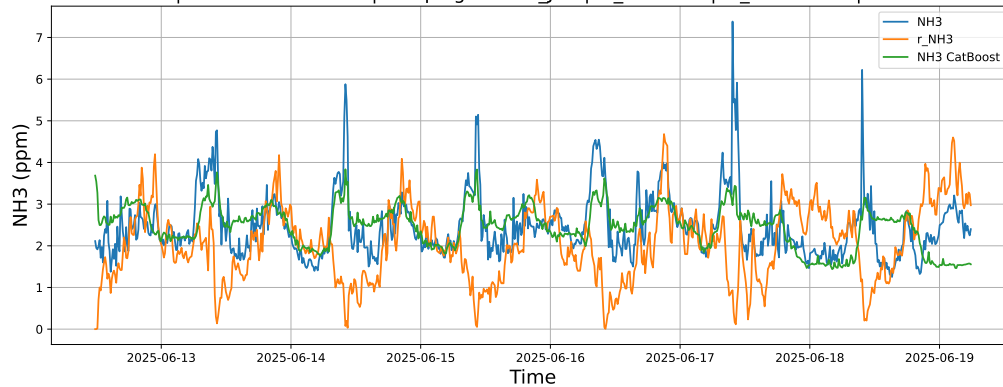
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



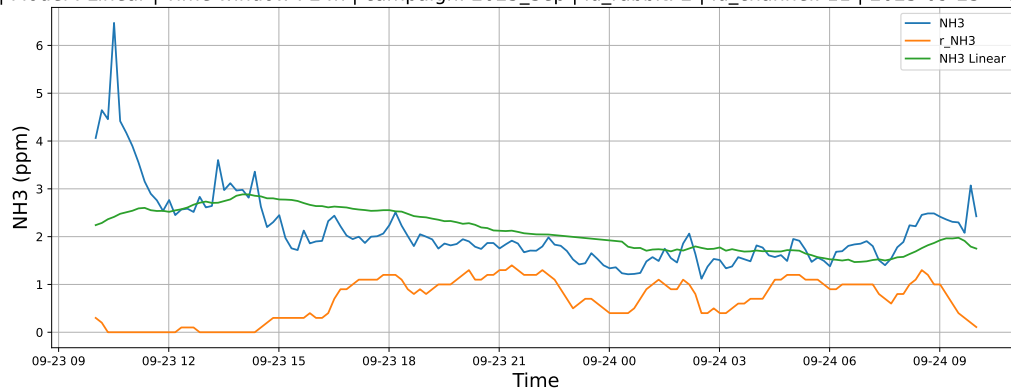
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



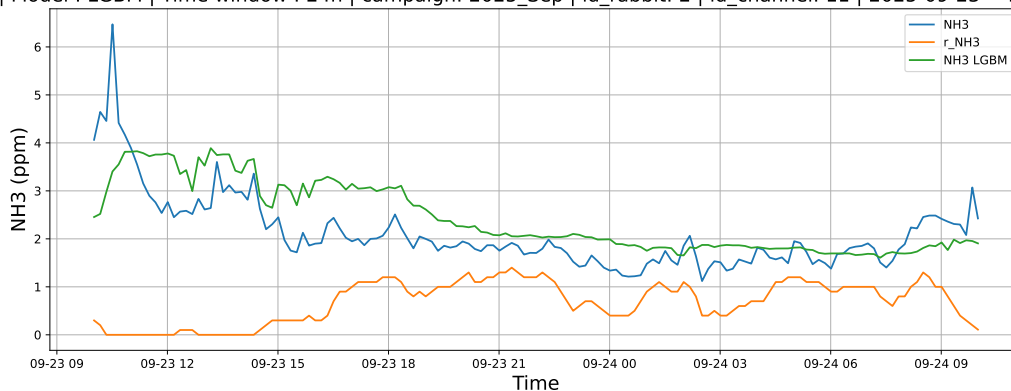
3.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

3.1.5.1 Time window : 24

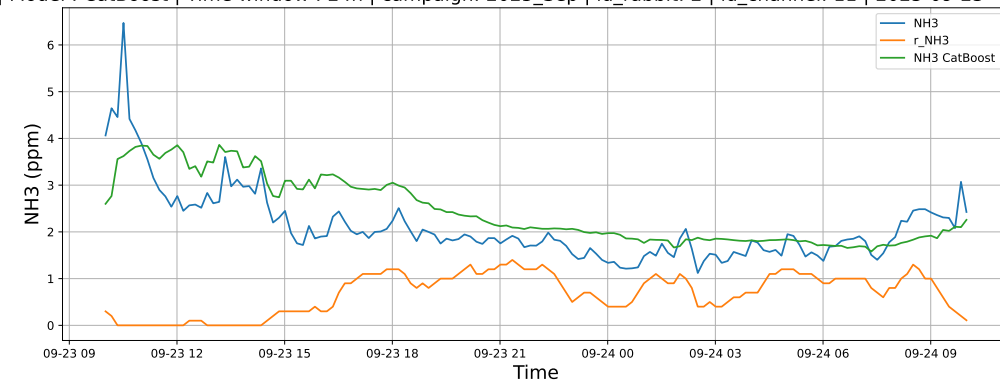
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

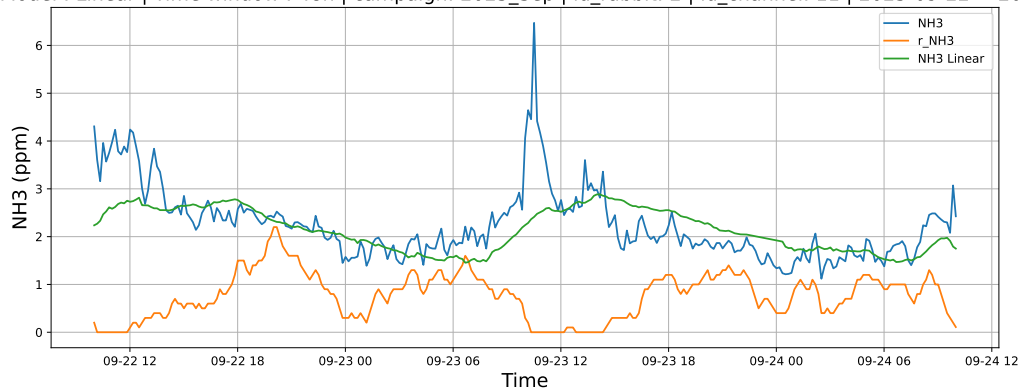


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

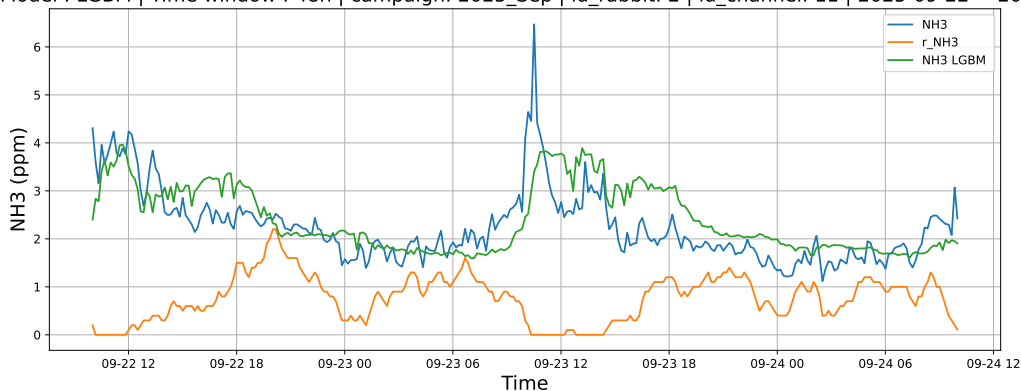


3.1.5.2 Time window : 48

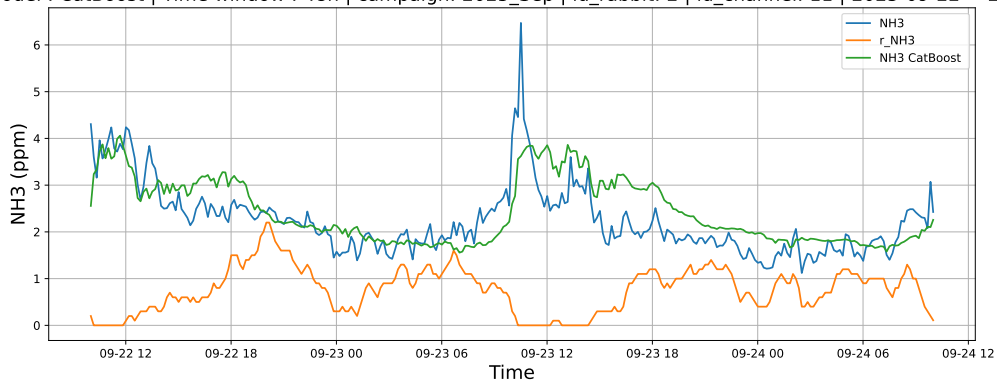
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

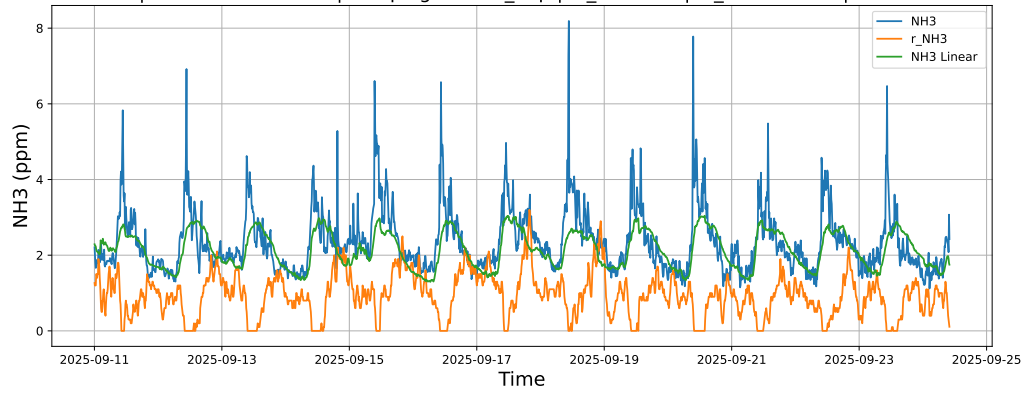


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

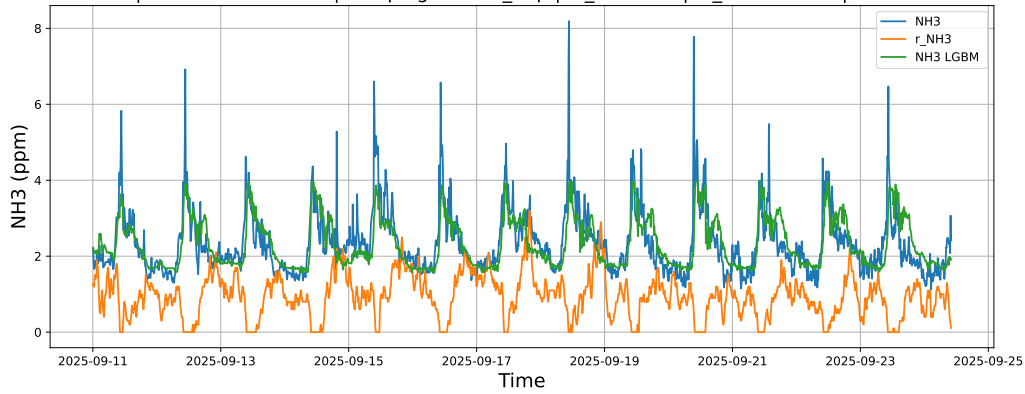


3.1.5.3 Time window : ALL

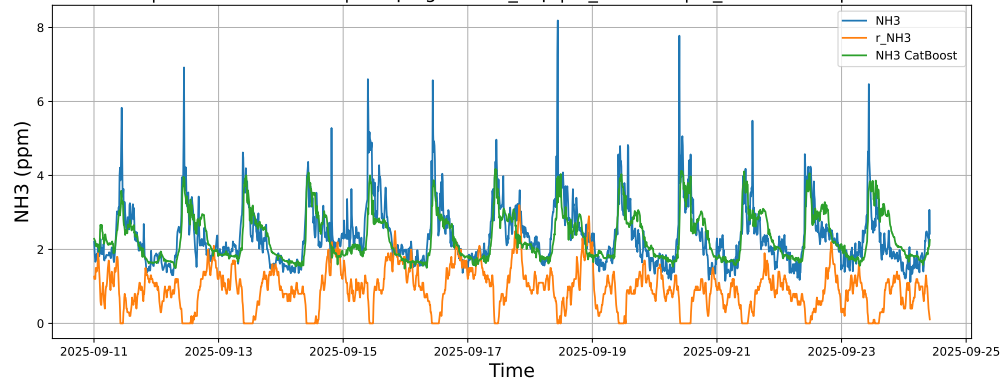
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



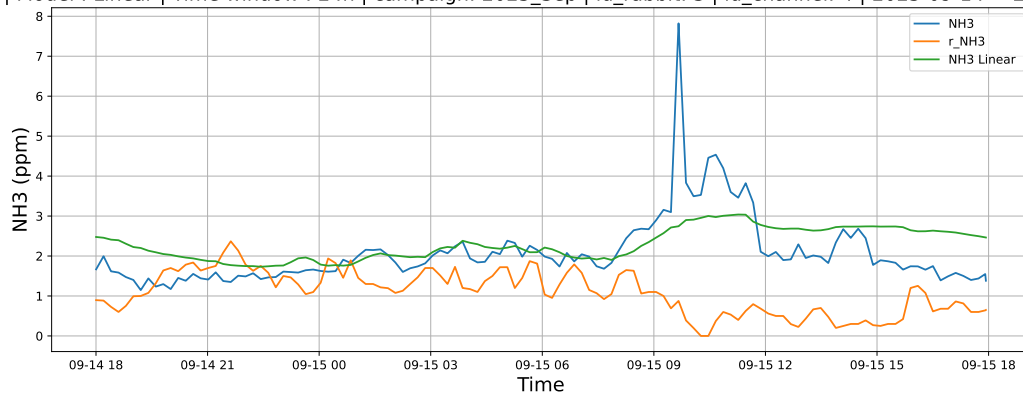
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



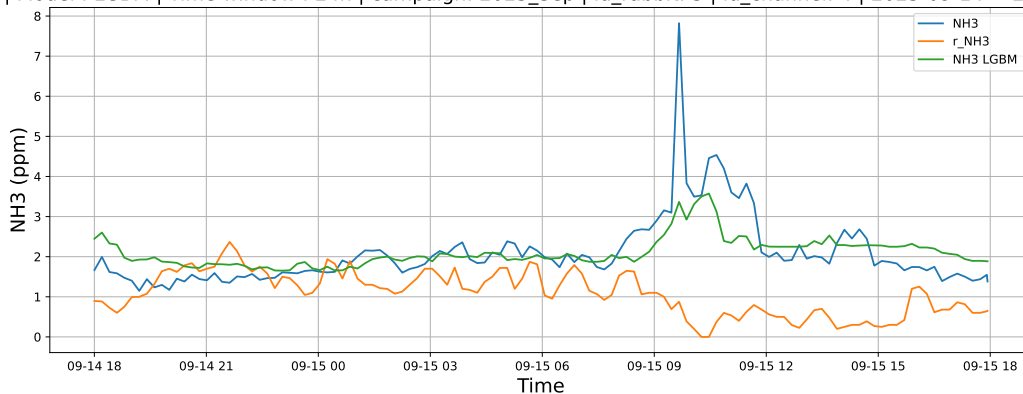
3.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

3.1.6.1 Time window : 24

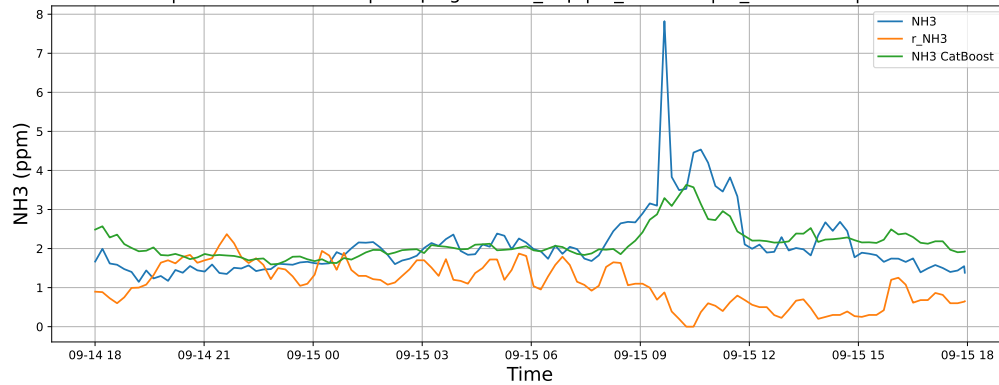
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

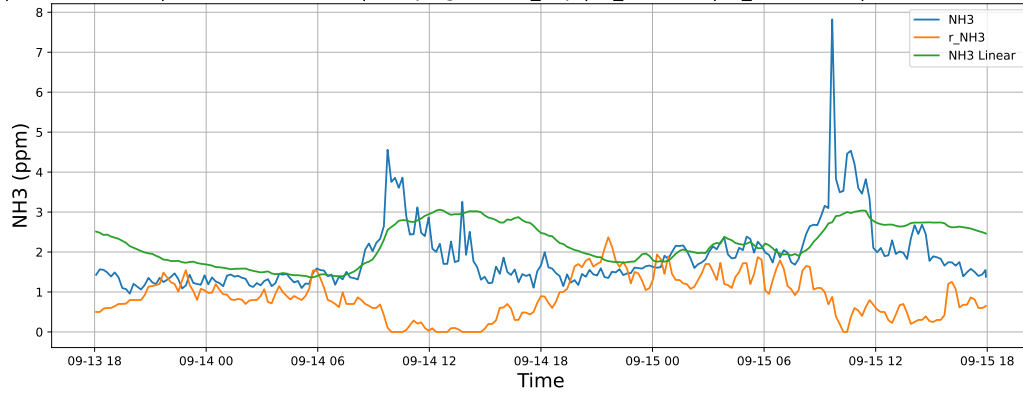


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

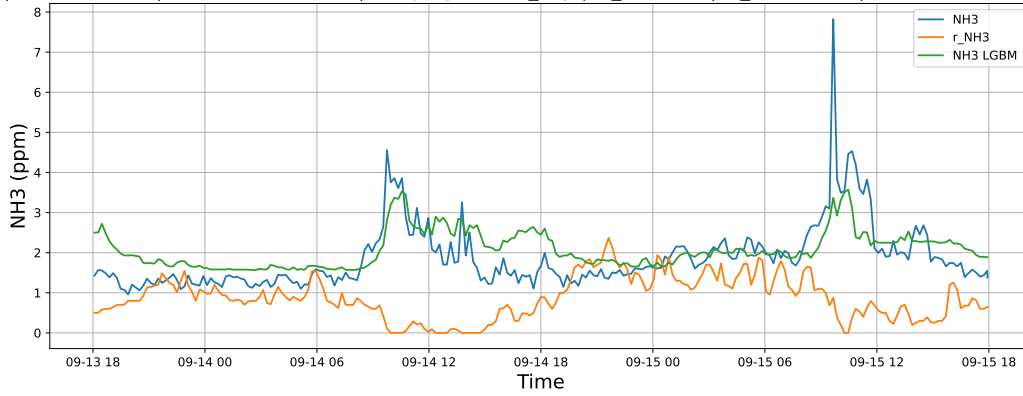


3.1.6.2 Time window : 48

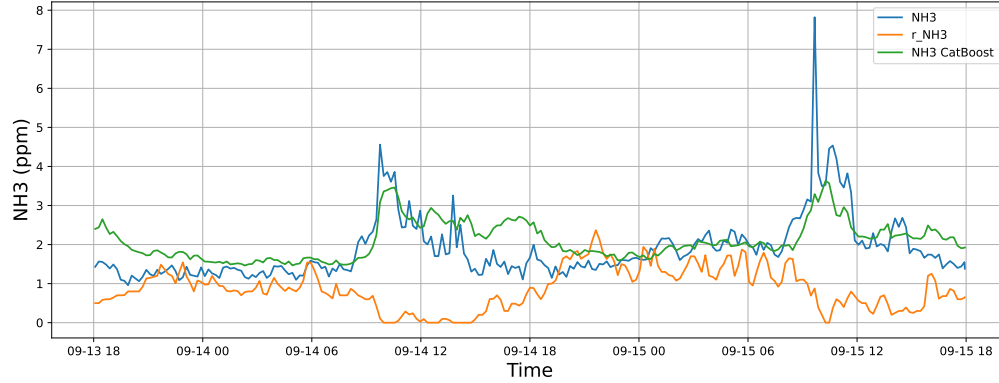
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

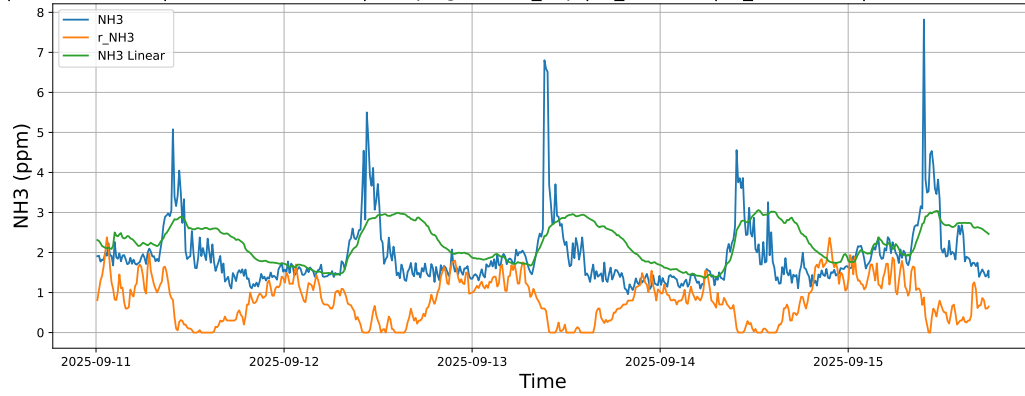


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

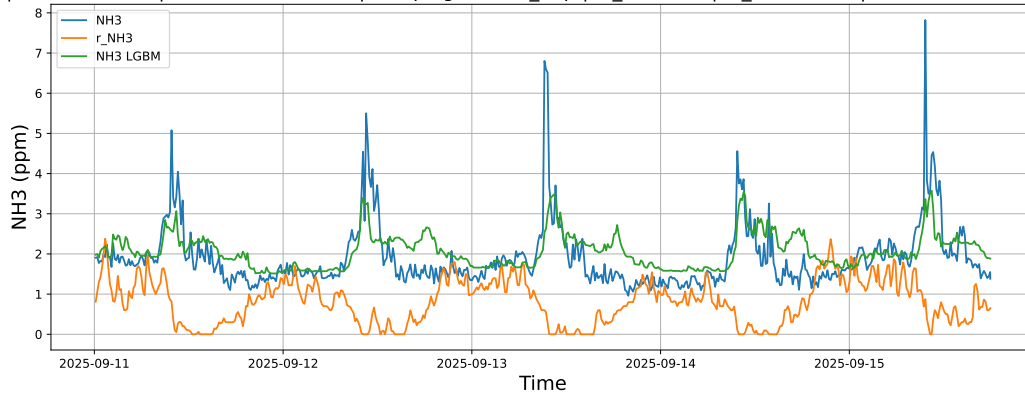


3.1.6.3 Time window : ALL

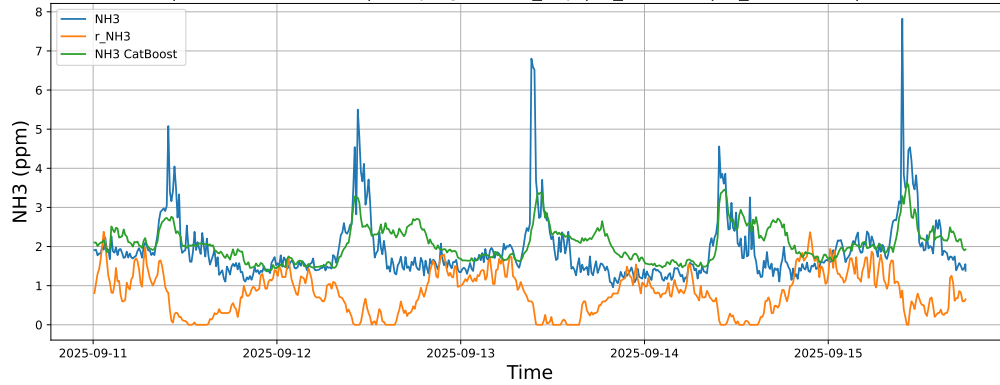
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



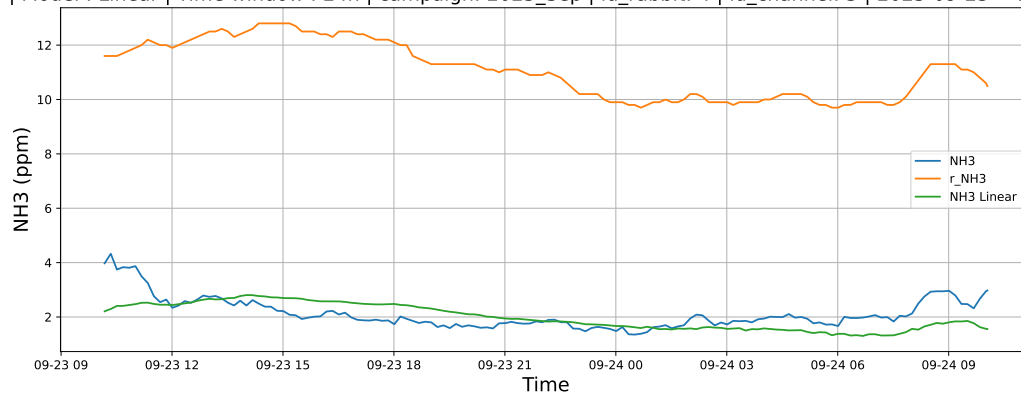
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



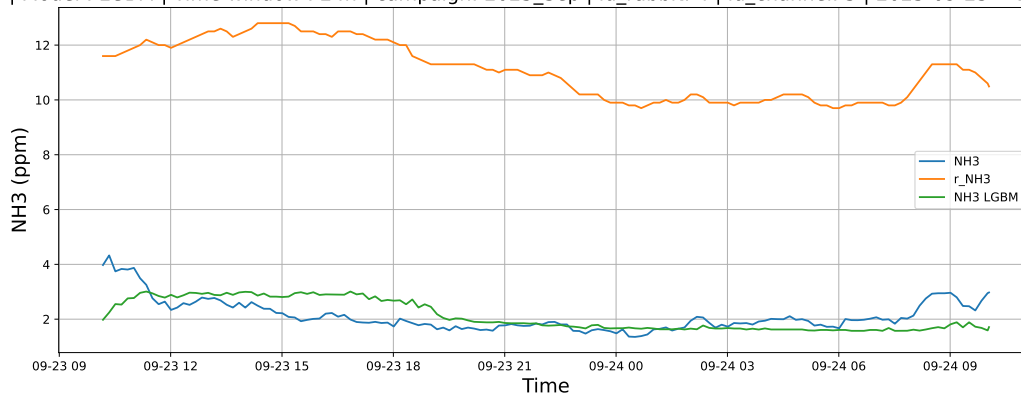
3.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

3.1.7.1 Time window : 24

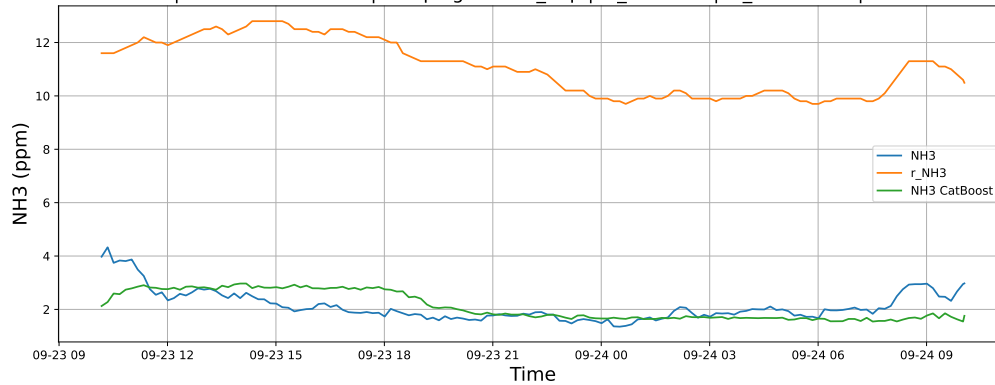
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

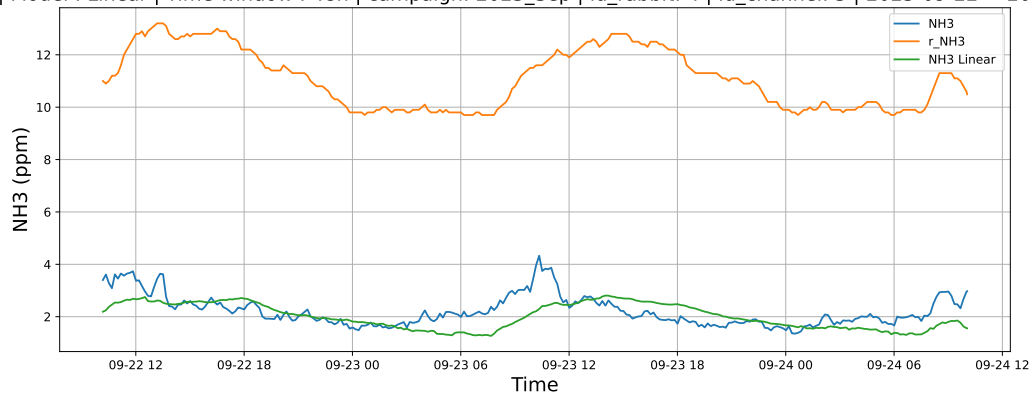


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

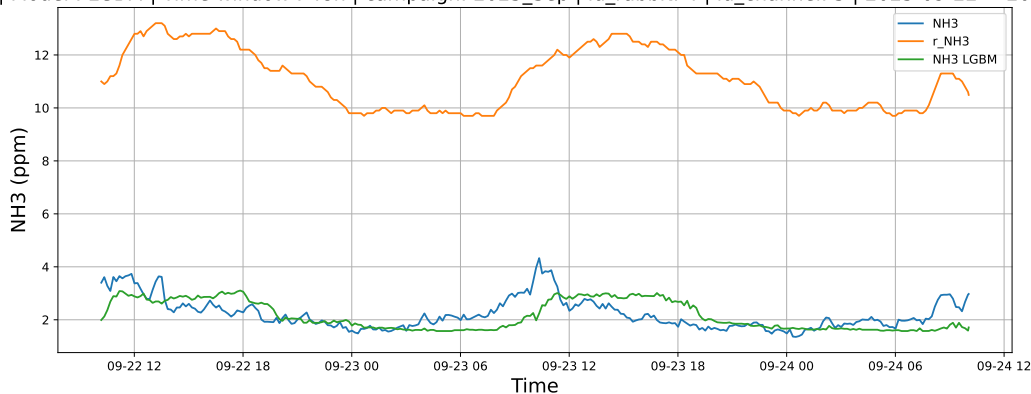


3.1.7.2 Time window : 48

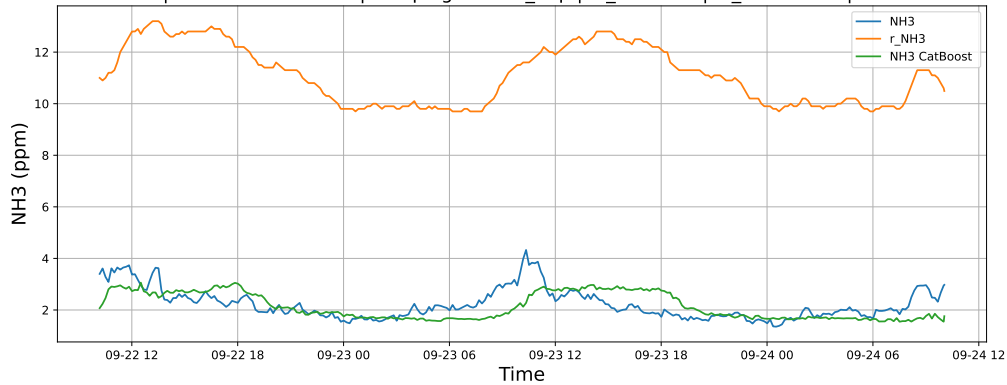
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

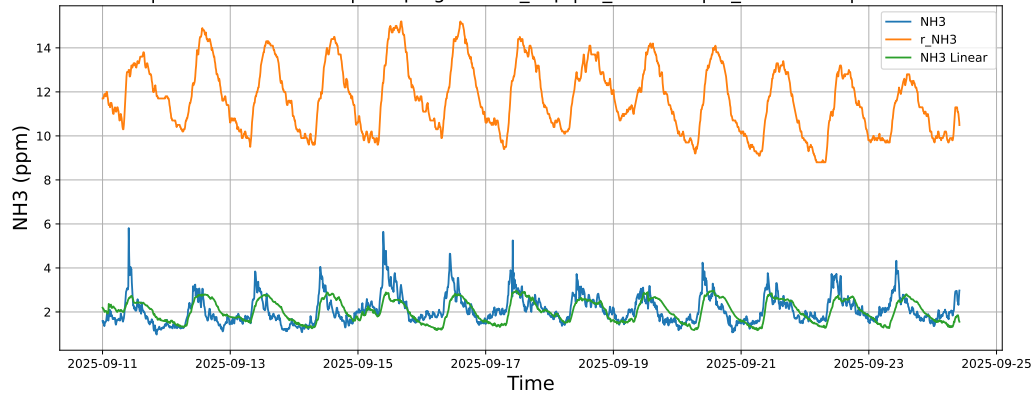


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

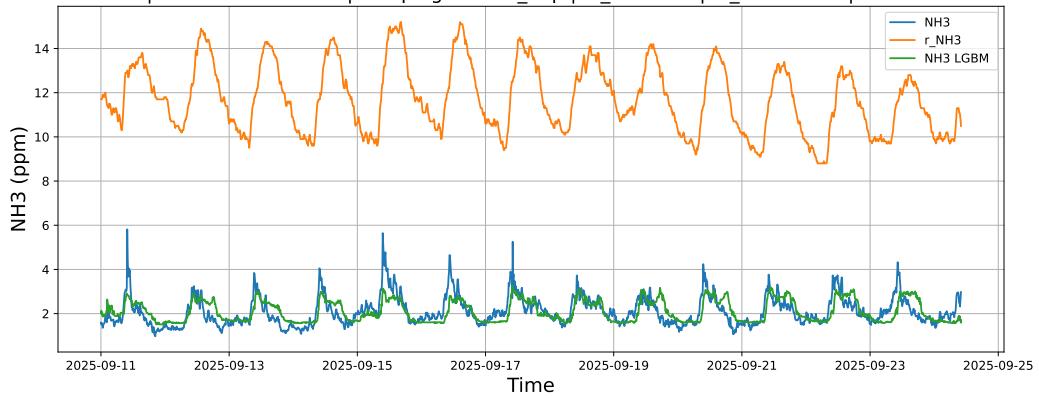


3.1.7.3 Time window : ALL

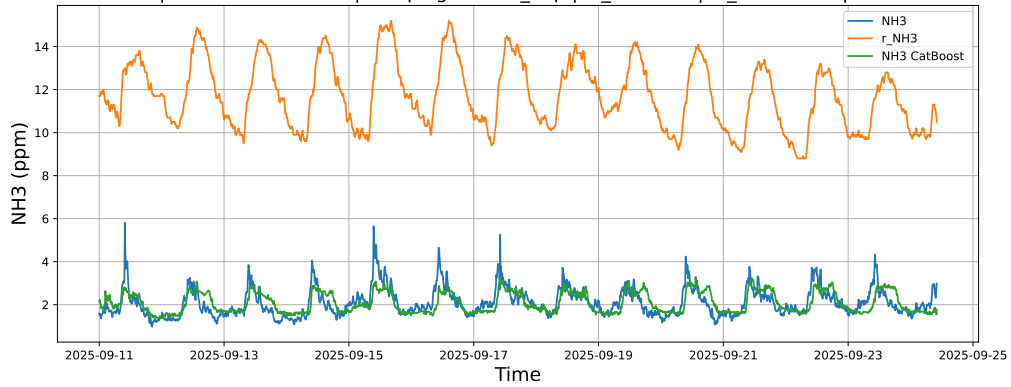
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



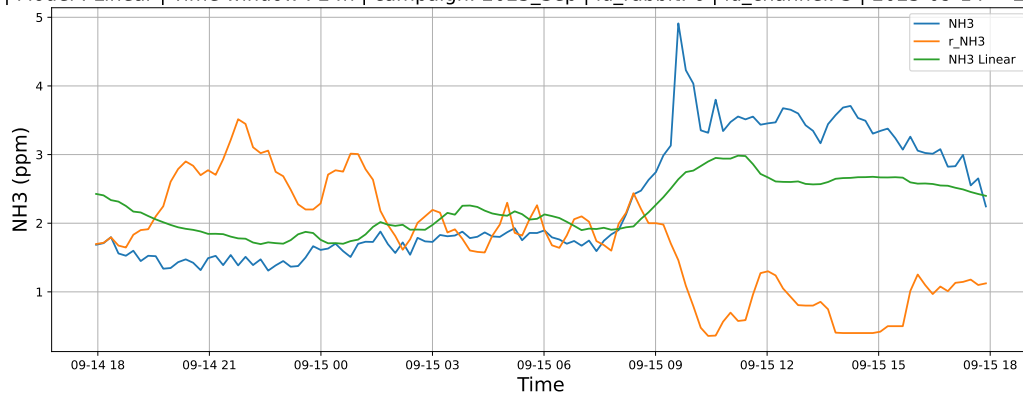
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



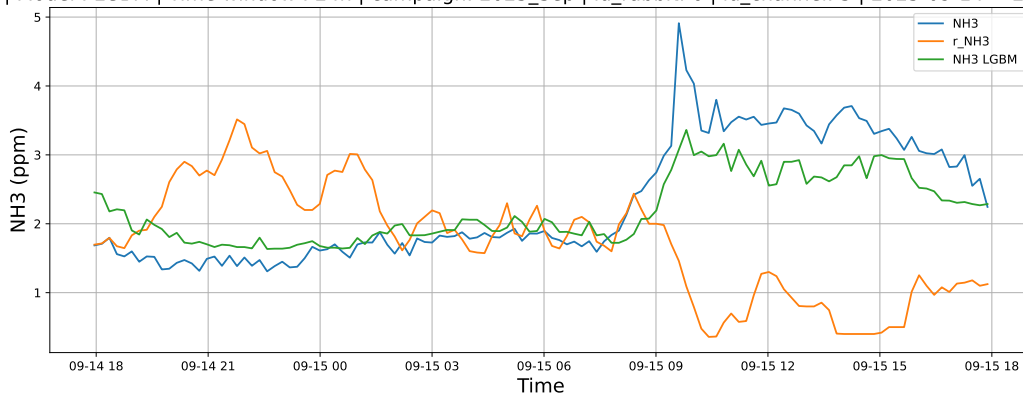
3.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.1.8.1 Time window : 24

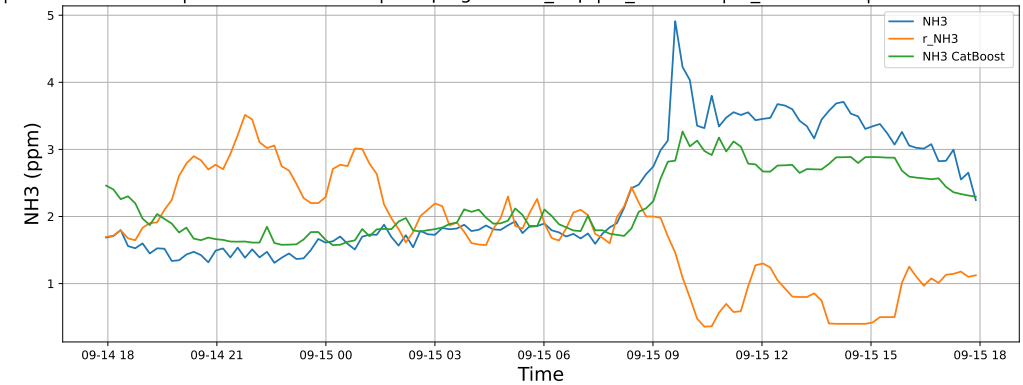
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

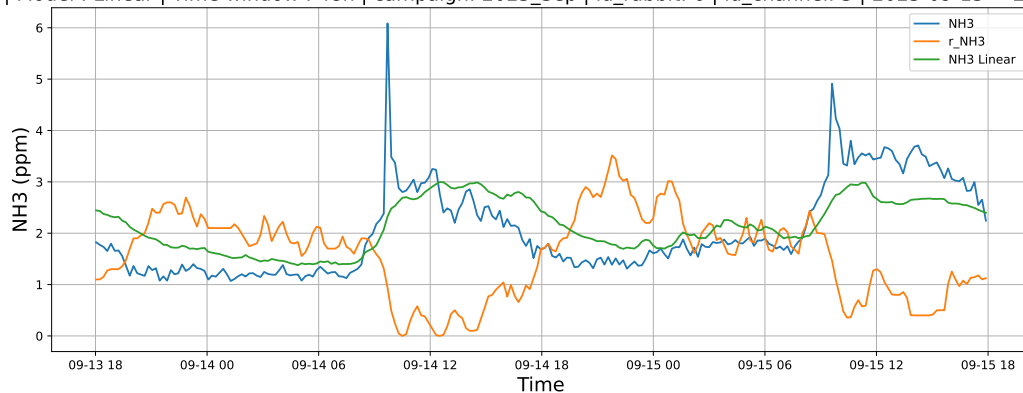


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

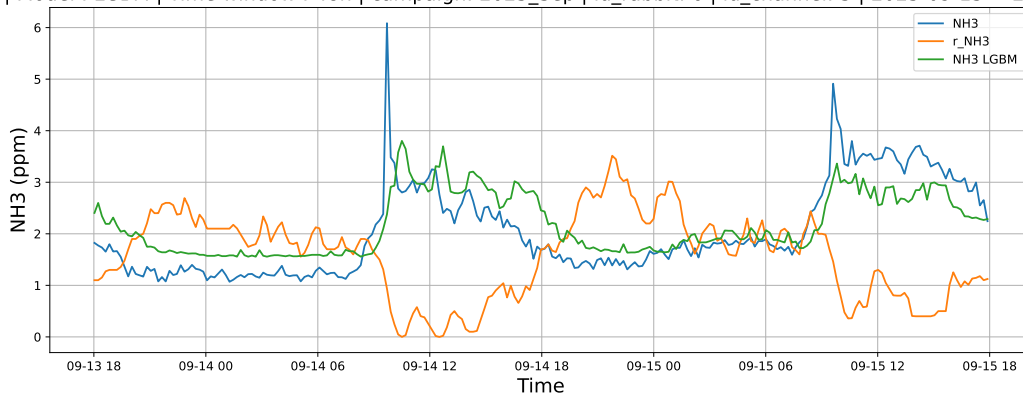


3.1.8.2 Time window : 48

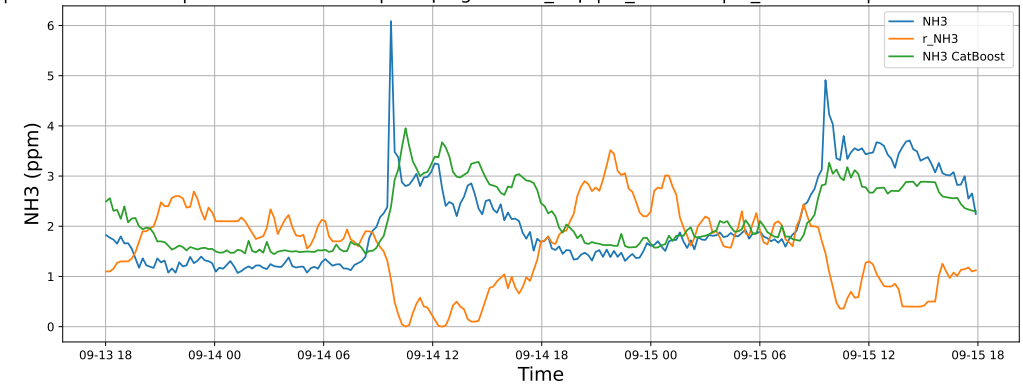
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

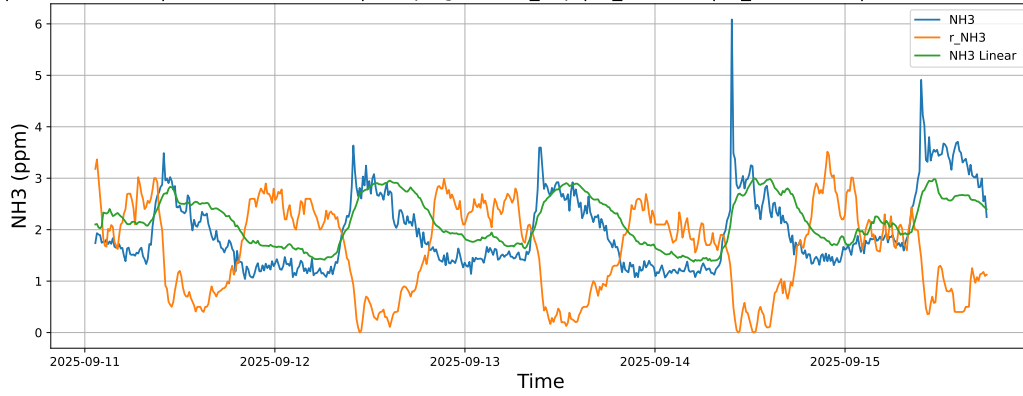


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

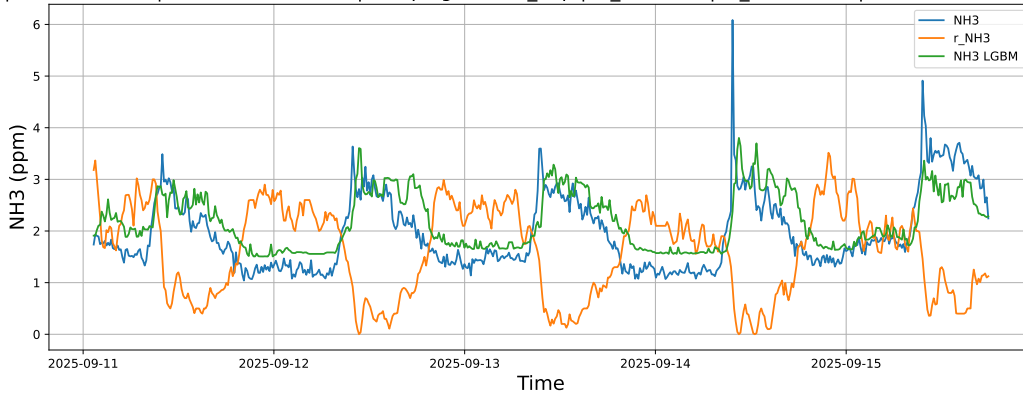


3.1.8.3 Time window : ALL

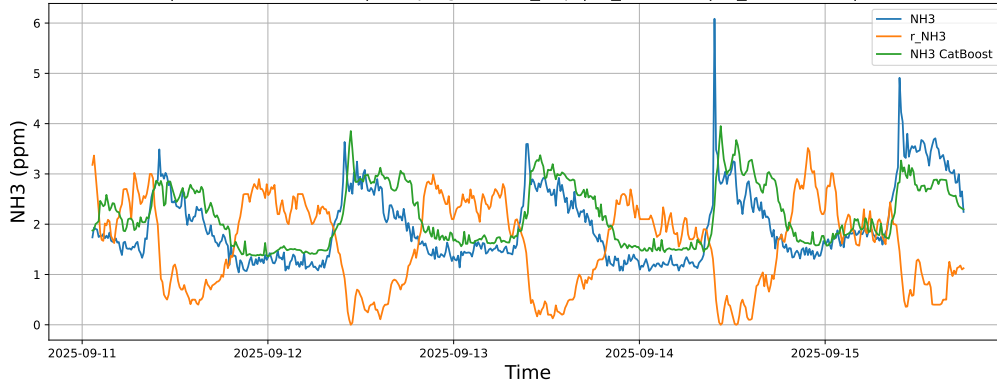
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

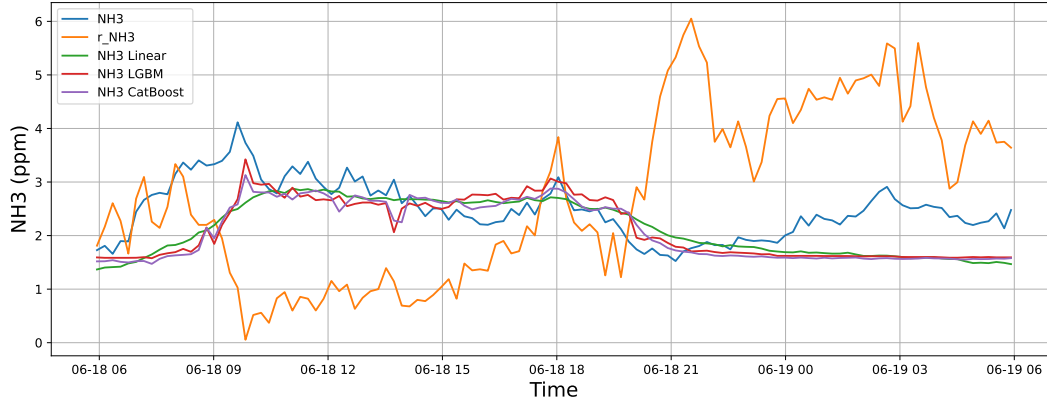


3.2 Compare plots

3.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

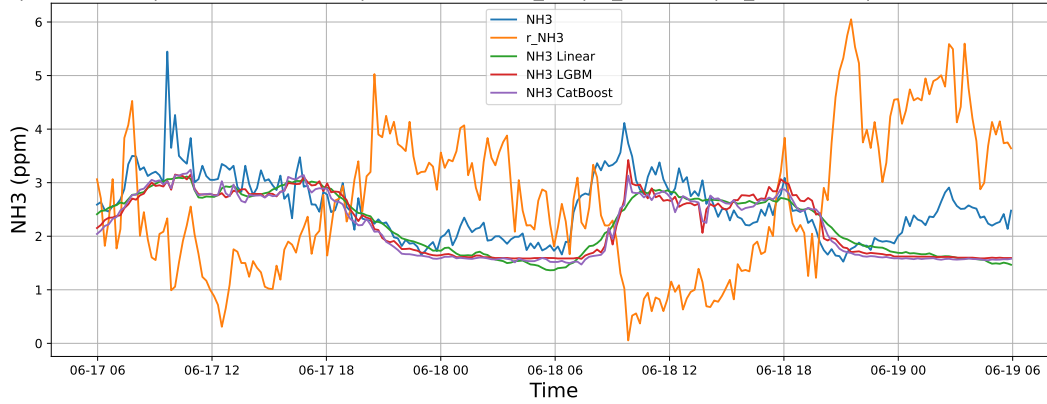
3.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



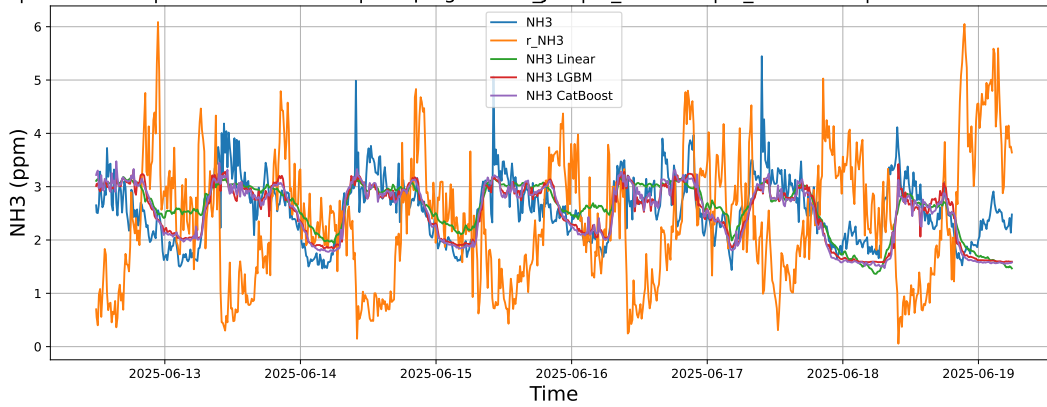
3.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



3.2.1.3 Time window : ALL

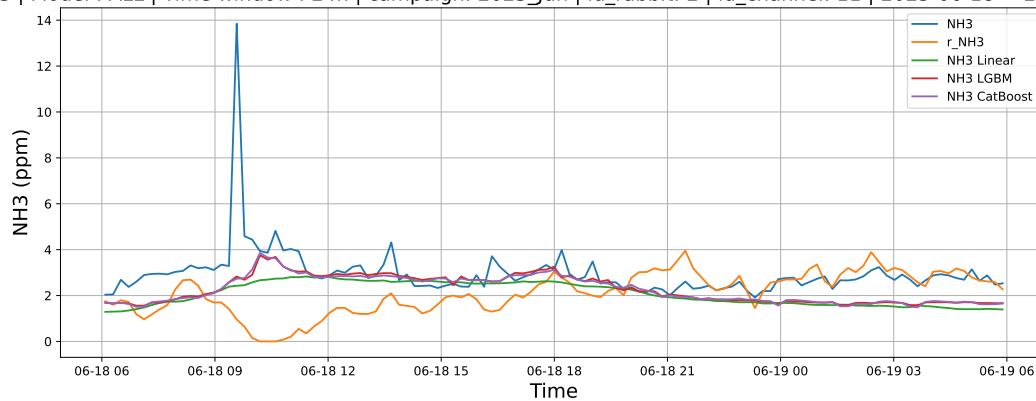
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



3.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

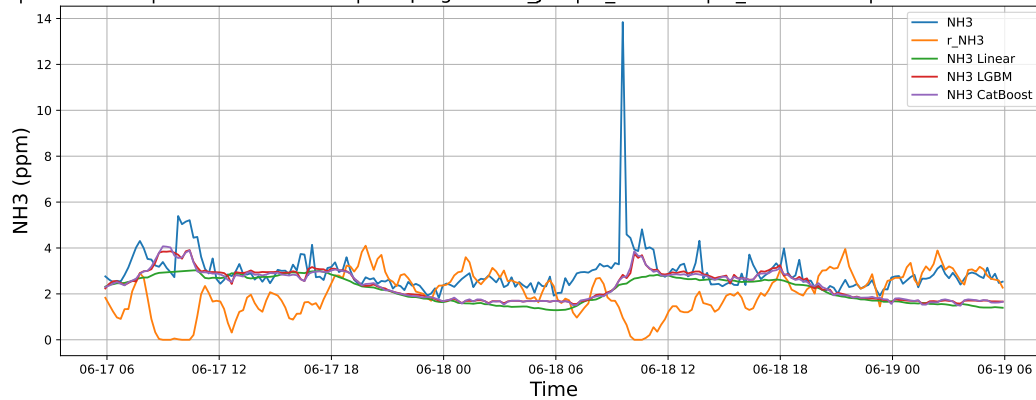
3.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



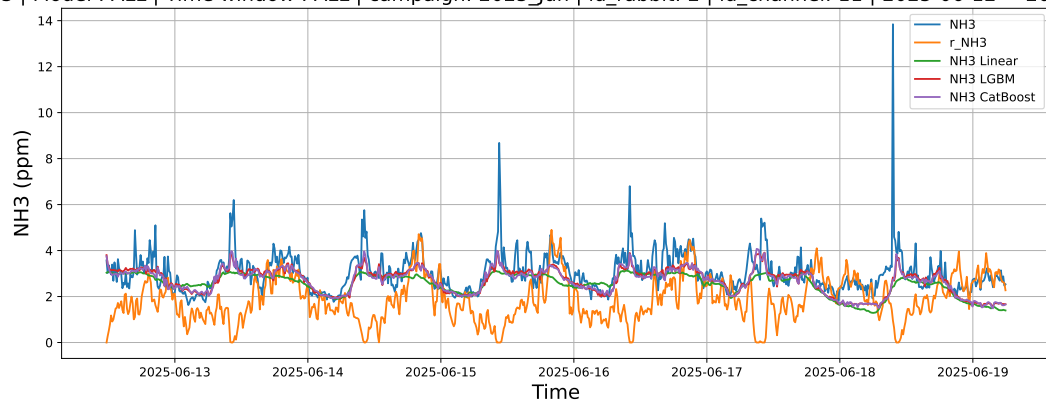
3.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



3.2.2.3 Time window : ALL

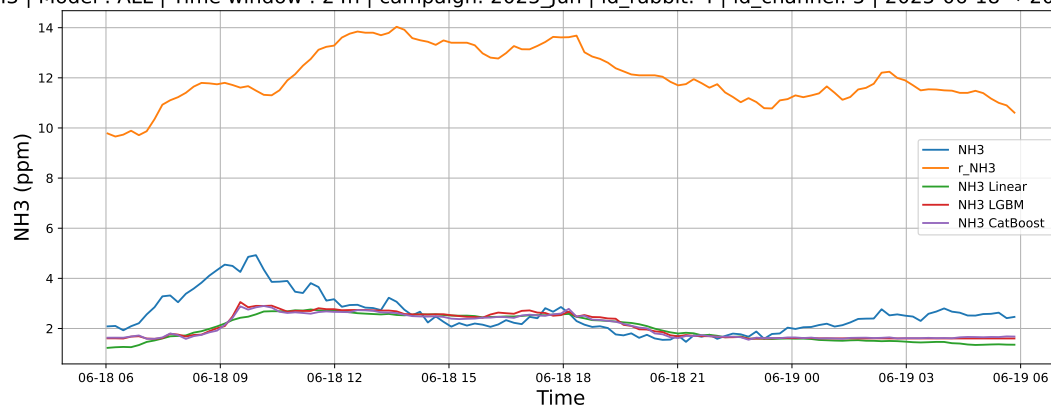
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



3.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

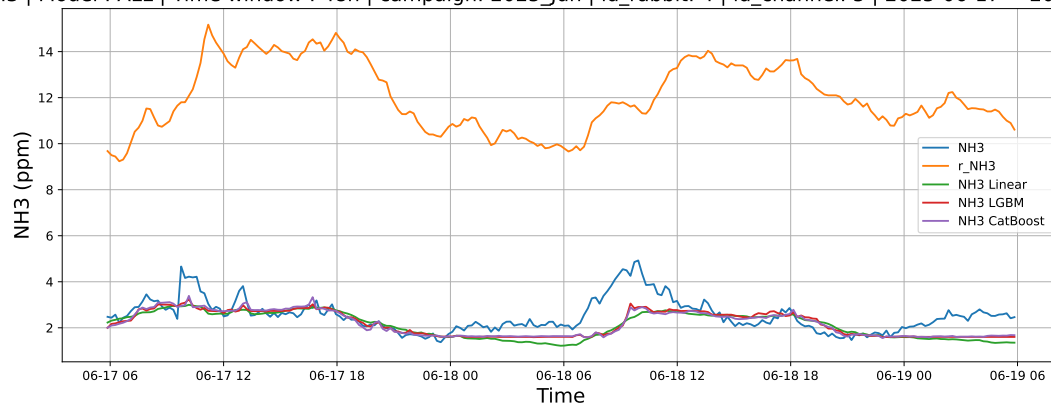
3.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



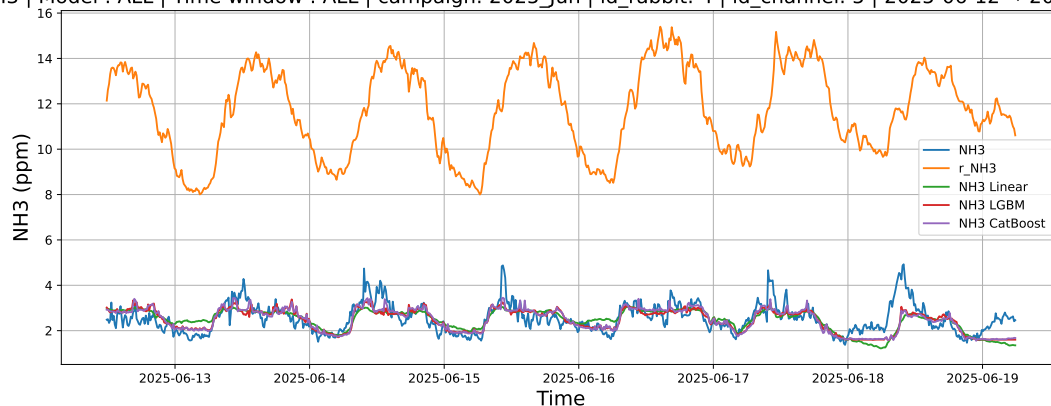
3.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



3.2.3.3 Time window : ALL

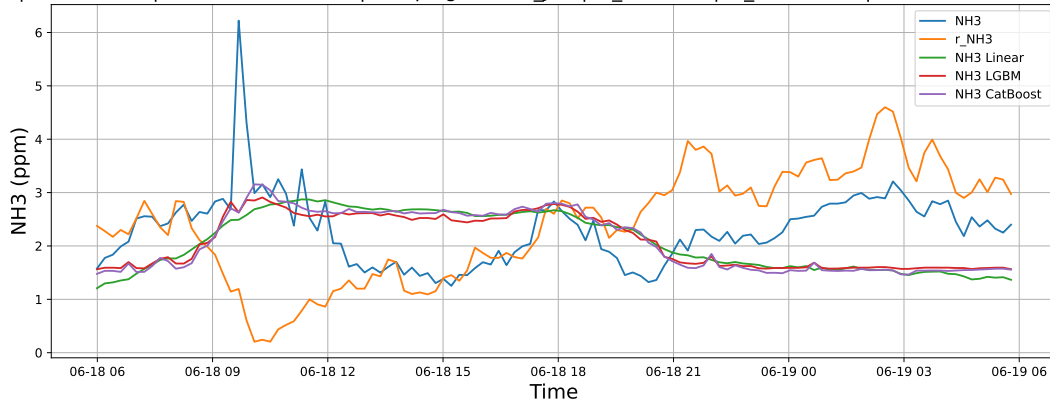
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



3.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

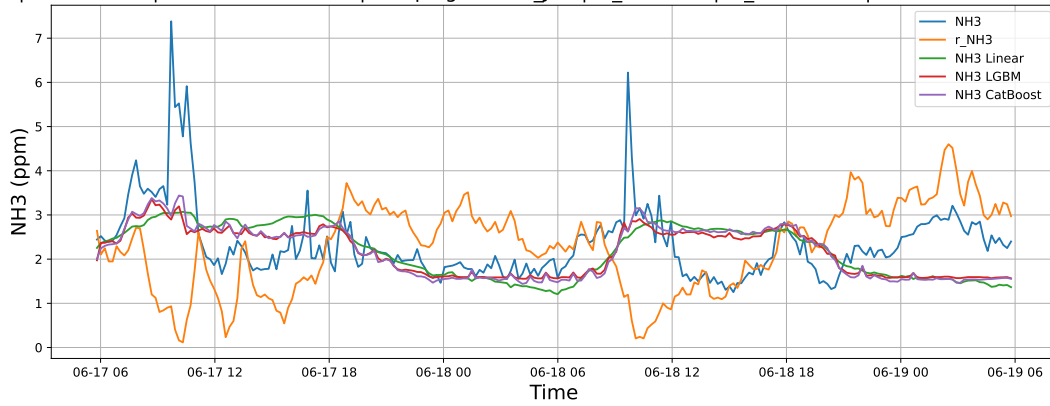
3.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



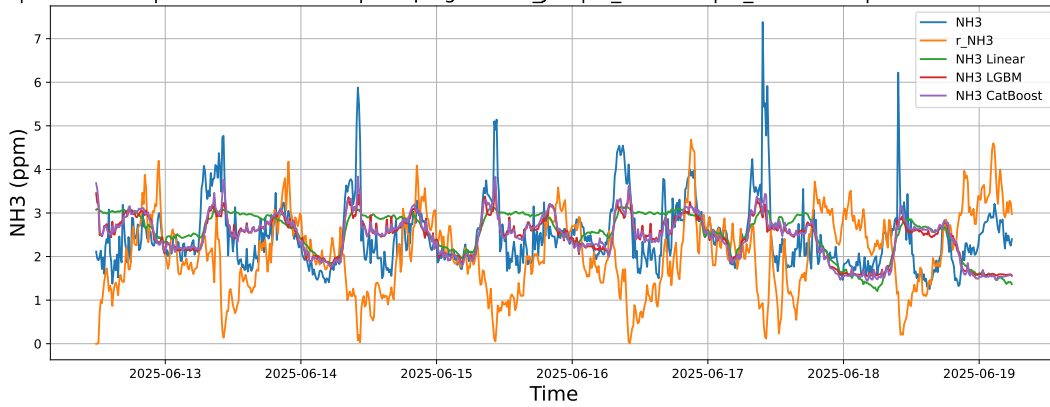
3.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



3.2.4.3 Time window : ALL

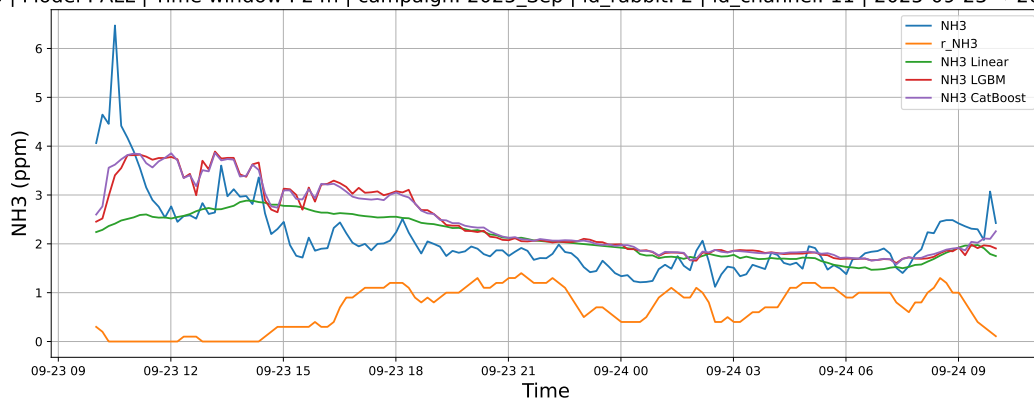
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



3.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

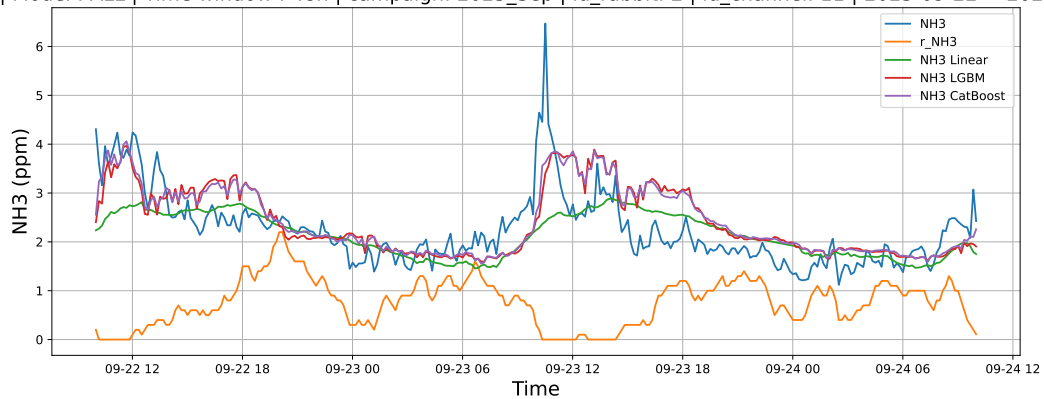
3.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



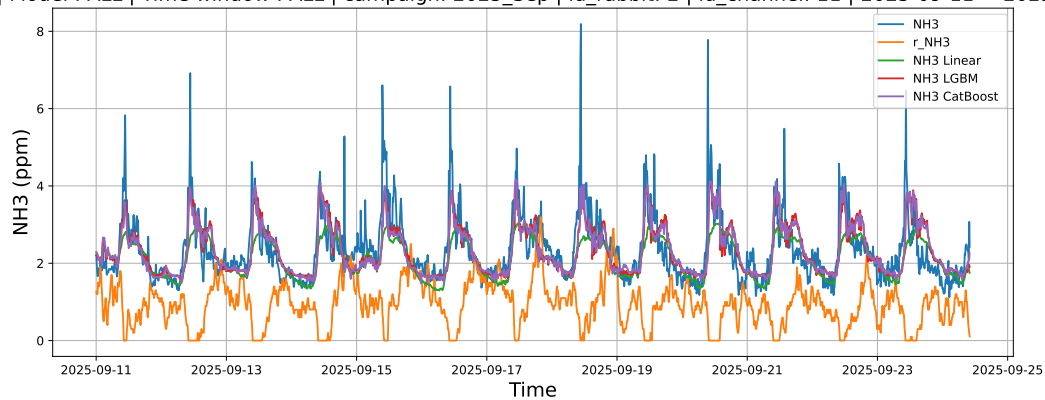
3.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



3.2.5.3 Time window : ALL

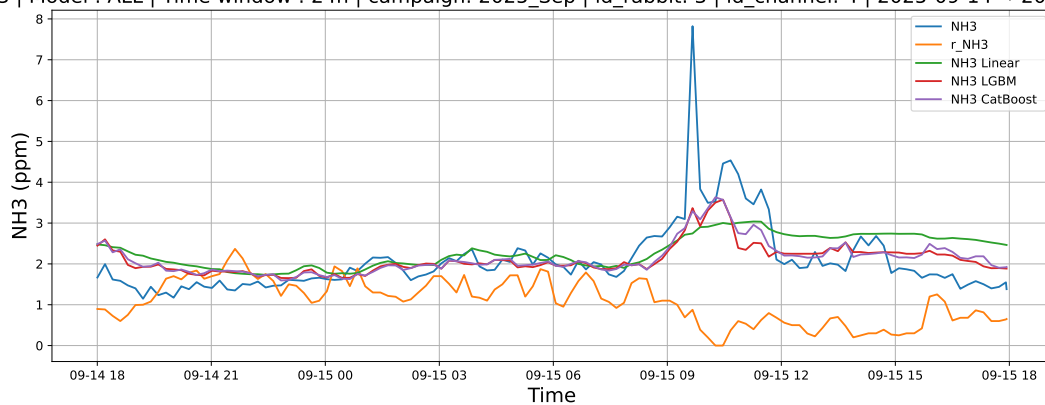
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



3.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

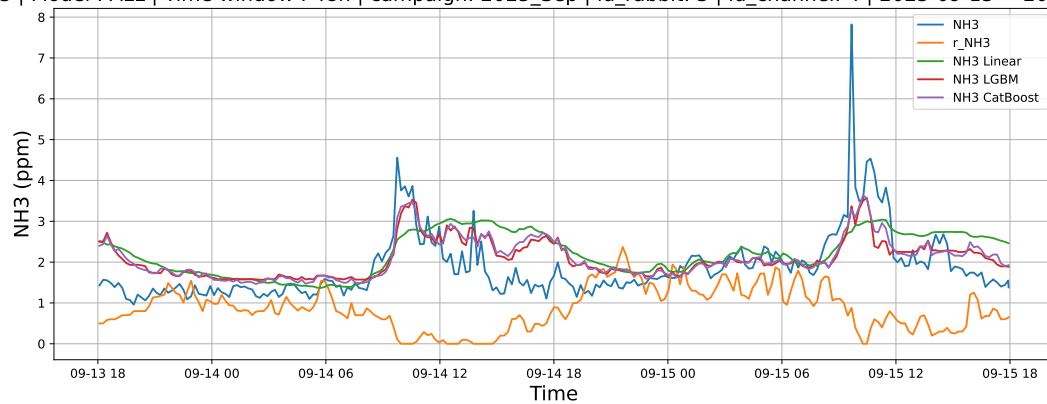
3.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



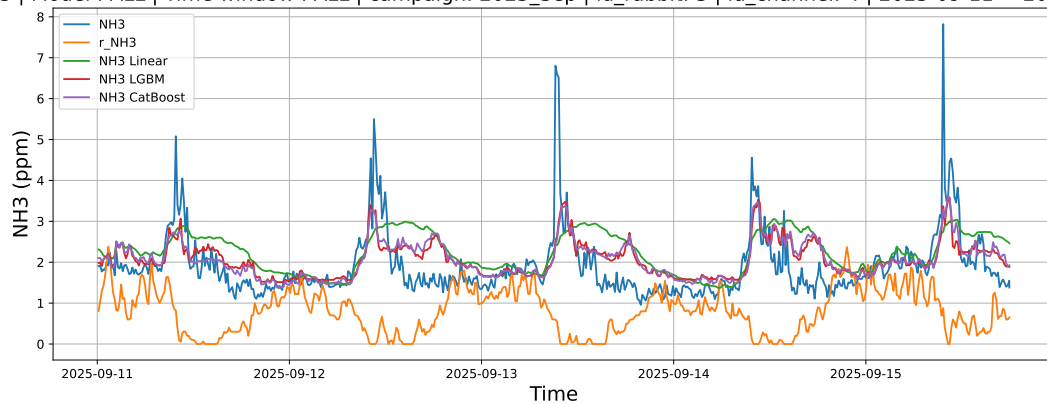
3.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



3.2.6.3 Time window : ALL

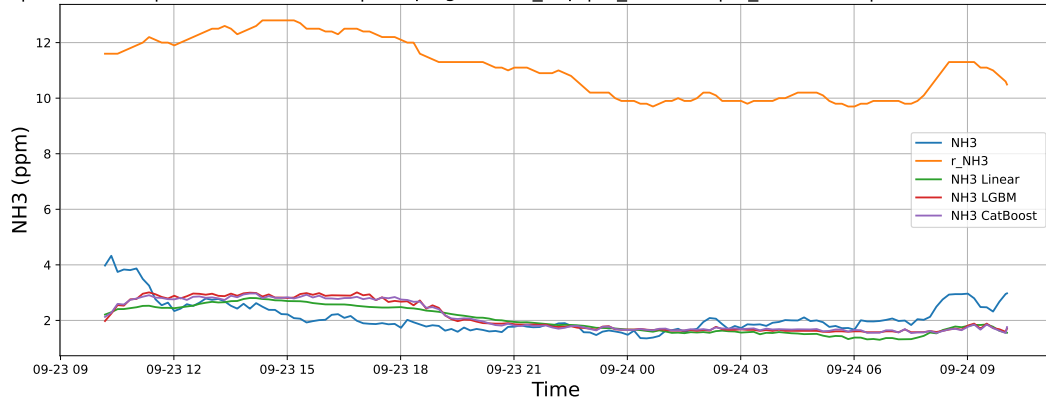
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



3.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

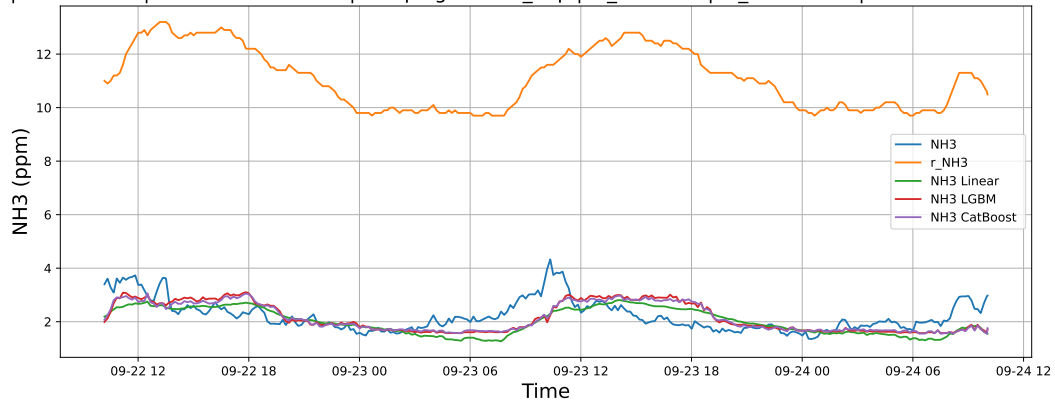
3.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



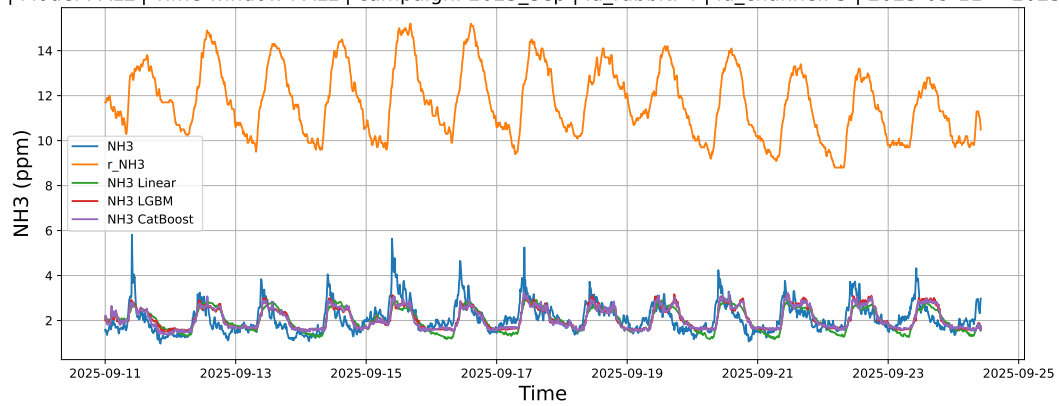
3.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



3.2.7.3 Time window : ALL

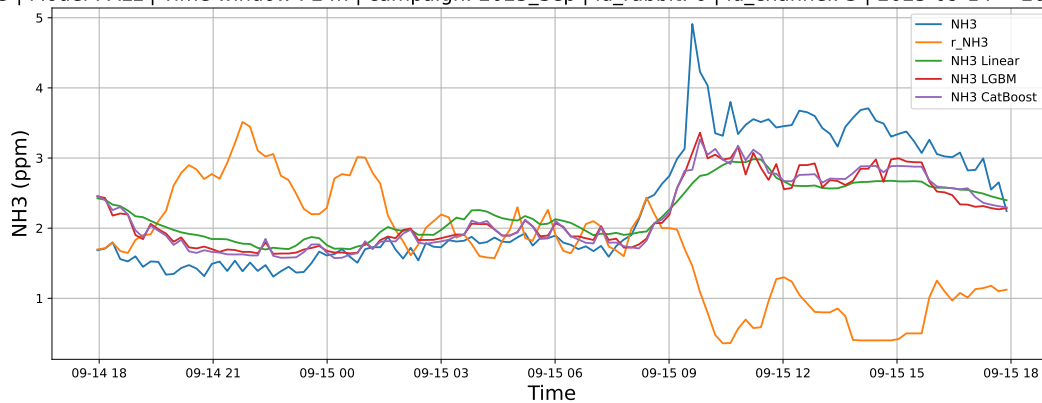
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



3.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

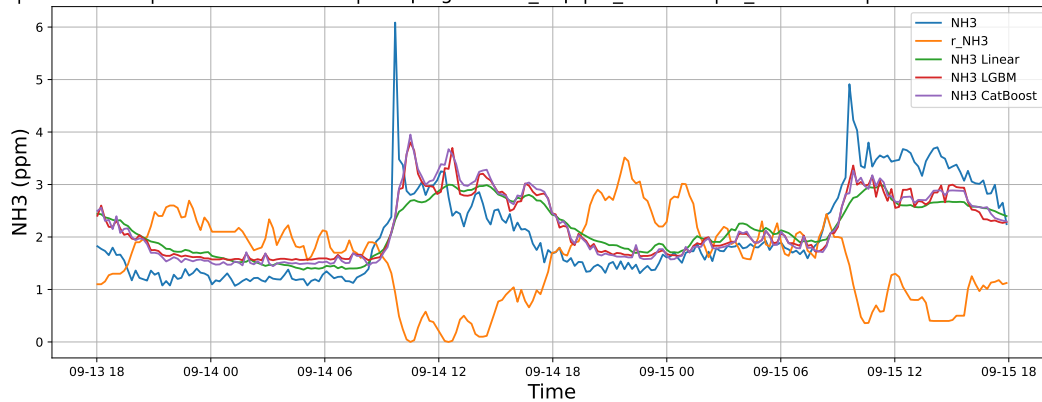
3.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



3.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

